








Time Series Analysis and Forecasting

Chapter 5b: Multivariate GARCH Models



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Learning Objectives

By the end of this chapter, you will be able to:

1. Explain why **multivariate volatility modeling** is necessary
2. Formulate and estimate **VEC**, **BEKK**, **CCC**, and **DCC** models
3. Interpret **dynamic correlations** and covariance decompositions
4. Apply multivariate GARCH for **portfolio optimization**, **hedging**, and **risk management**

Practical Skills

- ⊡ Python implementation with `rmgarch` / `mgarch` packages
- ⊡ Interpreting time-varying covariance matrices
- ⊡ Dynamic hedge ratios and portfolio weights

Key takeaway: *You are not learning models — you are learning how to measure real portfolio risk.*

Outline

Foundations

- Motivation: From Univariate to Multivariate
- The Multivariate GARCH Framework
- VEC and Diagonal VEC Models
- The BEKK Model
- Constant Conditional Correlation (CCC)
- Dynamic Conditional Correlation (DCC)

Applications

- Other Multivariate GARCH Models
- Estimation and the Curse of Dimensionality
- Model Selection and Diagnostics
- Portfolio Optimization
- Dynamic Hedging
- Summary and Quiz

The Story of This Chapter: The Problem

Core question: *How do we model risk when assets move together — and that co-movement changes over time?*

The Problem

- Correlations between assets **change over time** — and they spike during crises
- Ignoring this leads to wrong portfolios, bad hedges, and underestimated risk

The Difficulty

- Modeling N assets requires $N(N+1)/2$ time-varying covariance elements
- The **curse of dimensionality** makes most approaches infeasible
- The covariance matrix must be **positive definite** at every time step

The Story of This Chapter: The Solutions

Roadmap: *Each model solves one problem but creates another — until DCC.*

From general to practical

- **VEC**: the most general model — but unusable in practice ($O(N^4)$ parameters)
- **BEKK**: guarantees positive definiteness — but does not scale beyond $N = 5$
- **CCC**: simple and fast — but correlations are *constant* (empirically wrong)
- **DCC**: the optimal compromise — dynamic correlations with **only 2 extra parameters**

Key takeaway: *This chapter is a journey from generality to practicality. DCC wins because it scales.*

Why Go Multivariate?

The Problem

- Univariate GARCH models each asset's volatility **in isolation**
- They ignore **co-movements** between assets:
  - How does the volatility of asset A spill over to asset B?
  - How do correlations between assets change over time?

But in reality...

- Markets are interconnected — shocks propagate across assets
- Financial decisions require the **full covariance matrix** H_t, not just individual variances

Key insight: *Risk is not the sum of individual risks.*

What We Model: The Covariance Matrix H_t

One sentence: H_t *is the "weather forecast" for risk — it updates daily as new information arrives.*

Setup

- Given N assets with return vector $r_t = (r_{1t}, \ldots, r_{Nt})'$, we need to model:

$$H_t = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$$

- the **time-varying conditional covariance matrix**
- H_t contains all variances **and** all pairwise covariances at time t

Why Modeling H_t Is Hard

One sentence: *The number of parameters explodes with portfolio size.*

Three simultaneous challenges

- H_t has $N(N+1)/2$ unique elements — for $N = 50$: **1,275 parameters** at each t
- Must be **positive definite** at every time step (otherwise: negative portfolio variance)
- Must capture **dynamic** correlations, not just volatilities

Key takeaway: *Every multivariate GARCH model is a different answer to this same trade-off: flexibility vs. feasibility.*

Application 1: Portfolio Allocation

Markowitz with Time-Varying Inputs

- Optimal weights: $w_t^* = \frac{H_t^{-1}\mu_t}{1'H_t^{-1}\mu_t}$
- Weights shift **daily** as H_t updates
- Static covariance matrices lead to **suboptimal allocations** in turbulent periods

The Problem

- If H_t is treated as constant $\Rightarrow$ portfolio weights are wrong during crises
- Rebalancing too slowly $\Rightarrow$ excess risk; too fast $\Rightarrow$ excess cost

Key takeaway: *Optimal portfolios are dynamic, not static.*

Application 2: Dynamic Hedging

Minimum-Variance Hedge Ratio

- Optimal hedge: $h_t^* = \frac{\sigma_{\mathbf{12},t}}{\sigma_{\mathbf{22},t}}$
- Hedge more when spot–futures **covariance rises**
- Requires **time-varying covariances**, not just variances

Why Constant Hedging Fails

- In crisis: covariance structure shifts rapidly
- A fixed hedge ratio **under-hedges** or **over-hedges** depending on regime

Key insight: *Protection depends on the relationship between assets, not on individual volatilities.*

Application 3: Risk Management (VaR)

Portfolio Value-at-Risk

- $\text{VaR}_t^p = -\text{w}'\boldsymbol{\mu}_t + z_\alpha \sqrt{\text{w}'\text{H}_t\text{w}}$
- Risk depends on the **full covariance matrix**, not just individual volatilities

The Critical Problem

- Ignoring time-varying correlations $\Rightarrow$ **underestimates risk** during crises
- Correlations increase **precisely when diversification is needed most**

Key takeaway: *Correlations rise exactly when you need diversification the most.*

Stylized Fact: Correlations Are Not Constant

Empirical Evidence

- Correlations between assets are **not constant** — they increase during market turmoil
- This phenomenon is known as **correlation breakdown** or **contagion**
- Example: S&P 500 – DAX correlation jumped from 0.5 to 0.9 during the 2008 crisis
- **Diversification fails precisely when it is needed most**

Key insight: *Correlations are regime-dependent — they spike during crises and compress during calm periods.*

Numerical Example: The Impact of Correlations on Risk

Balanced Portfolio: 50% Stocks, 50% Bonds

- $\sigma_1 = 20\%$ (stocks), $\sigma_2 = 5\%$ (bonds)
- Portfolio variance: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$

Regime	ρ	σ_p	VaR 99% (1 day)
Calm market	-0.20	9.22%	-2.14%
Normal market	$+0.10$	10.25%	-2.38%
Crisis (2008)	$+0.60$	11.73%	-2.73%

Interpretation: *Portfolio risk increases by* ***27%*** *solely from the correlation change — it is the synchronization of movements, not individual volatility, that drives risk.*

Why It Matters: Practical Implications

Constant-Correlation Models Are Dangerous

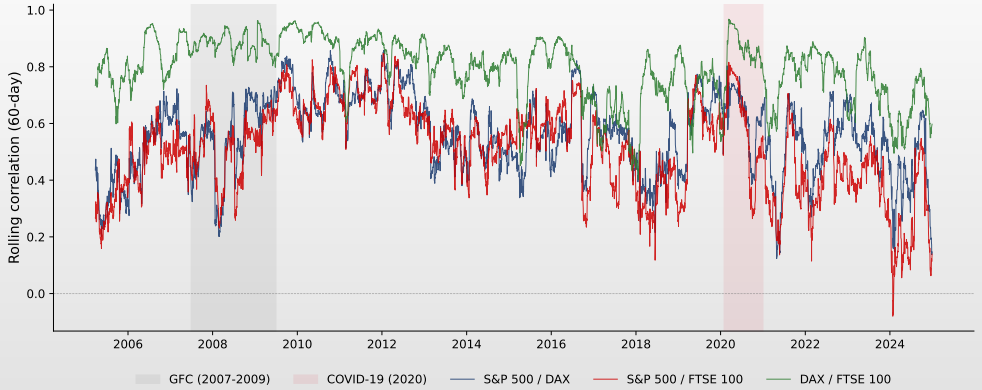
- **Portfolios**: weights based on stale correlations are suboptimal
- **Hedging**: fixed hedge ratios under-perform in regime shifts
- **VaR**: constant-ρ models underestimate tail risk during crises

Regulatory Implication

- Constant-correlation models are **inadequate** for:
  - Stress testing and scenario analysis
  - Real-time portfolio rebalancing
  - Regulatory capital calculations under Basel III/IV

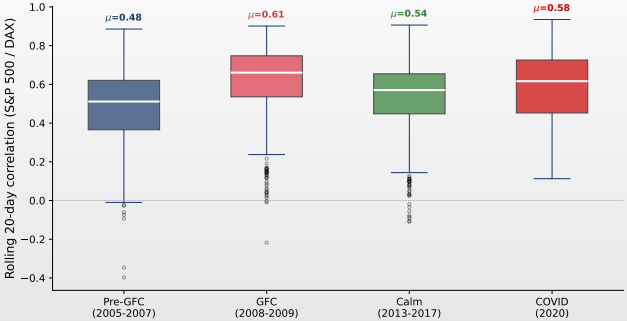
Key takeaway: *Models with constant correlations are dangerous in practice.*

Empirical Evidence: Rolling Correlations S&P 500, DAX, FTSE 100



TSA_ch5b_rolling_corr

Correlation Distribution: Crisis vs. Calm Regimes



Key Observations

- Median correlation increases significantly during crisis periods (GFC (*Global Financial Crisis*), COVID)
- Correlation dispersion rises — more uncertainty in cross-market relationships

TSA_ch5b_crisis_boxplot

Half-Vectorization — the vech(·) Operator

Definition

- The operator vech : $\mathbb{R}^{N\times N} \to \mathbb{R}^{N^*}$ stacks **column by column** the elements on and below the main diagonal of a symmetric matrix
- Output dimension: $N^* = N(N+1)/2$
- H_t is **symmetric**: $h_{ij,t} = h_{ji,t}$ — vech removes the $N(N-1)/2$ redundant elements

$$\mathrm{H}_t = \begin{pmatrix} h_{\mathbf{11},t} & h_{\mathbf{12},t} & h_{\mathbf{13},t} \\ h_{\mathbf{12},t} & h_{\mathbf{22},t} & h_{\mathbf{23},t} \\ h_{\mathbf{13},t} & h_{\mathbf{23},t} & h_{\mathbf{33},t} \end{pmatrix} \xrightarrow{\text{vech}} \begin{pmatrix} h_{\mathbf{11},t} \\ h_{\mathbf{12},t} \\ h_{\mathbf{13},t} \\ h_{\mathbf{22},t} \\ h_{\mathbf{23},t} \\ h_{\mathbf{33},t} \end{pmatrix}$$

- Elements in gray are duplicates — vech removes them.

Central Role in Multivariate GARCH

- Allows writing the model as a vector equation: $\mathrm{vech}(\mathrm{H}_t) = \mathrm{c} + \mathrm{A}\,\mathrm{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + \mathrm{B}\,\mathrm{vech}(\mathrm{H}_{t-1})$
- Dimension: $N^* = N(N+1)/2$ — for $N=2$: 3 elements; $N=10$: 55; $N=100$: 5,050

General Setup

Multivariate Return Process

- $\mathbf{r}_t = (r_{1t}, \ldots, r_{Nt})'$: N-dimensional return vector

$$\mathbf{r}_t = \boldsymbol{\mu}_t + \boldsymbol{\varepsilon}_t \tag{1}$$

$$\boldsymbol{\varepsilon}_t = \mathbf{H}_t^{1/2}\mathbf{z}_t, \qquad \mathbf{z}_t \overset{iid}{\sim} \mathcal{N}(0, \mathbf{I}_N) \tag{2}$$

- $\boldsymbol{\mu}_t$ = conditional mean vector (often a VAR or constant)
- $\mathbf{H}_t = \mathrm{Var}(\boldsymbol{\varepsilon}_t \mid \mathcal{F}_{t-1}) = N \times N$ conditional covariance matrix
- $\mathbf{H}_t^{1/2}$ = any matrix square root such that $\mathbf{H}_t^{1/2}(\mathbf{H}_t^{1/2})' = \mathbf{H}_t$

Intuition: $\mathbf{H}_t^{1/2}\mathbf{z}_t$ *transforms independent shocks into correlated returns — the "square root" imprints the dependence structure.*

The Challenge

- $\mathbf{H}_t$ has $N(N+1)/2$ unique elements. For $N = 50$: **1,275 time-varying parameters** at each t!

Positive Definiteness Requirement

Why Positive Definiteness Matters

- The conditional covariance matrix H_t must be **positive definite** (p.d.) for all t:
- Portfolio variance: $\sigma_p^2 = w' H_t w > 0$ for all $w \neq 0$
- The log-likelihood involves $\log|H_t|$ and H_t^{-1} — both require p.d.
- Cholesky decomposition for simulation requires p.d.

Strategies to Ensure Positive Definiteness

1. **Restrict parameters** so that H_t is p.d. by construction (BEKK)
2. **Decompose** $H_t = D_t R_t D_t$ and model D_t, R_t separately (DCC)
3. **Work in factor space** to reduce dimensionality (Factor GARCH, GO-GARCH)

The Cholesky Decomposition

Definition

- Any **symmetric positive definite** matrix H ($N \times N$) admits a **unique factorization**: $H = LL'$
- L is a **lower triangular** matrix with **strictly positive** diagonal elements
- Complexity: $O(N^3/3)$ — **twice as fast** as direct inversion

Example: the 2×2 case

- Let $H = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix}$. Then $L = \begin{pmatrix} \sqrt{h_{11}} & 0 \\ h_{12}/\sqrt{h_{11}} & \sqrt{h_{22} - h_{12}^2/h_{11}} \end{pmatrix}$

Role in Multivariate GARCH

- **Simulation**: $\varepsilon_t = L_t z_t$, where $z_t \sim N(0, I)$ — generates correlated shocks
- **Log-likelihood**: $\log |H_t| = 2 \sum_i \log L_{ii}$ — avoids direct determinant computation
- **Inversion**: solve $L_t x = \varepsilon_t$ by forward substitution ($O(N^2)$ vs. $O(N^3)$)

Parameter Proliferation: The Curse of Dimensionality

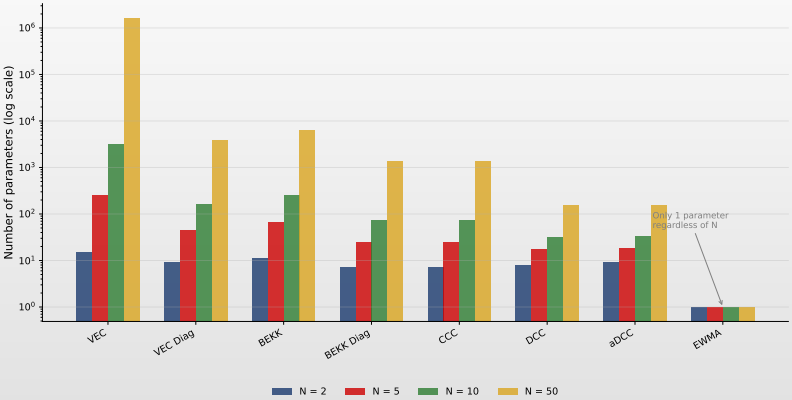
Number of Parameters by Model

Model	Parameters	$N = 5$
VEC(1,1)	$N(N+1)/2 + 2[N(N+1)/2]^2$	465
Diagonal VEC(1,1)	$3 \times N(N+1)/2$	45
BEKK(1,1)	$N(N+1)/2 + 2N^2$	65
Diagonal BEKK(1,1)	$N(N+1)/2 + 2N$	25
CCC(1,1)	$3N + N(N-1)/2$	25
DCC(1,1)	$3N + N(N-1)/2 + 2$	27

Practical Implication

- For large N, only **heavily parameterized models** (CCC, DCC, scalar BEKK, factor models) are feasible. Full VEC/BEKK become computationally intractable for $N > 5$.

Curse of Dimensionality: Visualization



TSA_ch5b_param_comparison

The Multivariate Normal Log-Likelihood

General Log-Likelihood

- For T observations of N-dimensional returns:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{TN}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\left[\underbrace{\log|\mathrm{H}_t|}_{\text{volume penalty}} + \underbrace{\boldsymbol{\varepsilon}_t'\mathrm{H}_t^{-1}\boldsymbol{\varepsilon}_t}_{\text{Mahalanobis distance}}\right]$$

Intuition: the likelihood penalizes both large covariance ("diffuse forecast") and large standardized residuals ("bad forecast").

Computational Challenges

- At each t: compute H_t ($O(N^2)$ or more), its inverse ($O(N^3)$), and determinant ($O(N^3)$)
- **Cholesky trick**: if $\mathrm{H}_t = \mathrm{L}_t\mathrm{L}_t'$, then $\log|\mathrm{H}_t| = 2\sum_i \log L_{ii}$ and solve $\mathrm{L}_t\mathrm{x} = \boldsymbol{\varepsilon}_t$ by forward substitution
- For DCC: two-step estimation dramatically reduces cost

The VEC Model — "Mother" of All MGARCH Models

: the most general linear MGARCH

- Every model you will learn (BEKK, CCC, DCC) is a **restricted version** of VEC
- The VEC(1,1) equation:

$$\text{vech}(H_t) = \underbrace{c}_{\text{intercept}} + \underbrace{A\,\text{vech}(\varepsilon_{t-1}\varepsilon_{t-1}')}_{\text{shock impact}} + \underbrace{B\,\text{vech}(H_{t-1})}_{\text{persistence}}$$

- c: $N^* \times 1$; A, B: $N^* \times N^*$; $N^* = N(N+1)/2$
- Full A allows every shock to affect every covariance element

Intuition: today's covariance = baseline + yesterday's shocks + yesterday's covariance.

Diagonal VEC: a Tractable Restriction

- $\text{vech}(H_t) = c + a \odot \text{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + b \odot \text{vech}(H_{t-1})$ ($\odot$ = element-wise)
- Each variance/covariance follows its own GARCH — **no cross-effects**
- Reduces parameters from $O(N^4)$ to $3N^*$

The Hadamard Product

Definition

- Let $\mathrm{A}, \mathrm{B} \in \mathbb{R}^{m \times n}$. The **Hadamard product** $\mathrm{A} \odot \mathrm{B}$ is the **element-wise** multiplication:

$$(\mathrm{A} \odot \mathrm{B})_{ij} = a_{ij} \cdot b_{ij}$$

- Matrices must have the **same dimensions** — the result preserves dimension $m \times n$

Example: the 2×2 case

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \odot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 & 2 \cdot 6 \\ 3 \cdot 7 & 4 \cdot 8 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 21 & 32 \end{pmatrix}$$

Role in Multivariate GARCH

- Diagonal VEC: $\mathrm{vech}(\mathrm{H}_t) = \mathrm{c} + \mathrm{a} \odot \mathrm{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + \mathrm{b} \odot \mathrm{vech}(\mathrm{H}_{t-1})$
- Each element of H_t gets its **own coefficients** a_k, b_k — no cross-interactions. Reduces parameters from N^{*2} (full VEC) to $3N^*$ (Diagonal VEC)

The Diagonal VEC Model

Definition (Bollerslev, Engle & Wooldridge, 1988)

- Restrict A and B to be **diagonal**: $\text{vech}(H_t) = c + a \odot \text{vech}(\varepsilon_{t-1}\varepsilon'_{t-1}) + b \odot \text{vech}(H_{t-1})$
- $\odot$ = Hadamard product; a, b: vectors $N^* \times 1$

Element by element for $N = 2$

$$h_{11,t} = c_{11} + a_{11}\,\varepsilon^2_{1,t-1} + b_{11}\,h_{11,t-1} \tag{3}$$

$$h_{12,t} = c_{12} + a_{12}\,\varepsilon_{1,t-1}\varepsilon_{2,t-1} + b_{12}\,h_{12,t-1} \tag{4}$$

$$h_{22,t} = c_{22} + a_{22}\,\varepsilon^2_{2,t-1} + b_{22}\,h_{22,t-1} \tag{5}$$

Intuition: each variance/covariance follows its own GARCH — the shock of asset 2 cannot affect the variance of asset 1.

Trade-off

- $3N^*$ parameters (manageable), but each element evolves **independently** — no cross-asset propagation

VEC: Bivariate Example

Example: two assets ($N = 2$)

- With full VEC, $N^* = 3$:

$$\begin{pmatrix} h_{11,t} \\ h_{12,t} \\ h_{22,t} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1}^2 \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{2,t-1}^2 \end{pmatrix} + \mathrm{B} \begin{pmatrix} h_{11,t-1} \\ h_{12,t-1} \\ h_{22,t-1} \end{pmatrix}$$

Interpretation

- Off-diagonal elements of A capture **shock spillovers between assets**
  - a_{13}: the impact of asset 2's shock on the variance of asset 1
- Off-diagonal elements of B capture **volatility persistence spillovers**
- Total parameters: $3 + 9 + 9 = 21$ (for just 2 assets!)

VEC: Key Insight and Common Pitfalls

: VEC is the "mother" of all MGARCH models

- Every multivariate GARCH model (BEKK, CCC, DCC) is a **restricted version** of VEC
- The VEC equation is the most general linear dynamics for H_t
- All restrictions are choices about which parameters to eliminate

: confusing VEC and Diagonal VEC

- Diagonal VEC (A, B diagonal) eliminates *all cross-effects*
- Each $h_{ij,t}$ depends only on its own lag and on $\varepsilon_{i,t-1}\varepsilon_{j,t-1}$
- It does **not** just mean "fewer parameters": it fundamentally eliminates volatility propagation

- **Never** use full VEC for $N > 2$ — $O(N^4)$ parameters, estimation diverges
- **Avoid** Diagonal VEC when spillovers matter (e.g., banking contagion studies)
- **Skip** VEC if you only need portfolio variance — CCC/DCC are faster and guaranteed p.d.

VEC Section Summary

3 Things to Remember

1. VEC is the most **general** linear MGARCH model — all other models restrict it
2. Diagonal VEC is estimable but eliminates propagation; full VEC is rich but infeasible for $N > 2$
3. The central unresolved problem: **no guarantee of positive definiteness**

Model Positioning

- **Use when**: $N = 2$ and maximum flexibility is needed
- **vs. BEKK**: BEKK guarantees p.d. with same expressiveness — prefer BEKK

From VEC to BEKK: What Went Wrong?

One sentence: *VEC has no built-in mechanism to ensure* H_t *stays positive definite.*

VEC's two fatal flaws

- VEC is the most **general** multivariate GARCH — every element of H_t has its own dynamics
- But: (1) **no guarantee** that H_t is positive definite, (2) parameter count **explodes** ($O(N^4)$)
- Diagonal VEC solves (2) but still fails at (1) — and loses spillover effects

The core problem

- VEC writes $\mathrm{vech}(\mathrm{H}_t) = \mathrm{c} + \mathrm{A}\,\mathrm{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + \mathrm{B}\,\mathrm{vech}(\mathrm{H}_{t-1})$
- Nothing in this equation **forces** H_t to be p.d. at every t
- Sufficient conditions exist (Engle & Kroner, 1995) but are **hard to impose**

BEKK's Insight: Build Positive Definiteness Into the Structure

One sentence: *Use quadratic forms — they are always positive semi-definite by construction.*

The key idea

- Write $H_t = CC' + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + B'H_{t-1}B$
- Each term is p.s.d. **by construction**: $X'MX \succeq 0$ for any $M \succeq 0$
- Sum of p.s.d. matrices + a p.d. matrix (CC') = p.d. always

Key takeaway: *BEKK solves VEC's biggest flaw by making positive definiteness automatic — no extra constraints needed.*

The BEKK Model (Engle & Kroner, 1995)

Definition: BEKK(1,1)

$$H_t = \underbrace{CC'}_{\text{intercept (p.d.)}} + \underbrace{A'\varepsilon_{t-1}\varepsilon'_{t-1}A}_{\text{shock impact (p.s.d.)}} + \underbrace{B'H_{t-1}B}_{\text{persistence (p.s.d.)}}$$

- C: $N \times N$ lower triangular; A, B: $N \times N$ matrices

Intuition: the quadratic form $A'MA$ is always p.s.d. — positive definiteness comes "for free."

Key Properties

- H_t is **positive definite by construction** (if CC' is p.d.)
- Captures both **own-volatility persistence** and **cross-volatility spillovers**
- Parameters: $N(N+1)/2 + 2N^2$ — for $N = 3$: $6 + 18 = 24$

▶ **Practical Warning:** BEKK does not scale — $2N^2$ params $\Rightarrow$ infeasible for $N > 5$. BEKK = Baba, Engle, Kraft, Kroner (1990).

***Key takeaway:** BEKK trades scalability for guaranteed positive definiteness and spillover identification.*

BEKK: Bivariate Case

Expanding the BEKK(1,1) for $N = 2$

- Let $\mathrm{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $\mathrm{B} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, $\mathrm{C} = \begin{pmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{pmatrix}$. Then:

$$h_{11,t} = c_{11}^2 + a_{11}^2 \varepsilon_{1,t-1}^2 + 2a_{11}a_{21}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{21}^2 \varepsilon_{2,t-1}^2 + b_{11}^2 h_{11,t-1} + 2b_{11}b_{21}h_{12,t-1} + b_{21}^2 h_{22,t-1} \tag{6}$$

$$h_{12,t} = c_{11}c_{21} + a_{11}a_{12}\varepsilon_{1,t-1}^2 + (a_{11}a_{22} + a_{12}a_{21})\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{21}a_{22}\varepsilon_{2,t-1}^2 + b_{11}b_{12}h_{11,t-1} + (b_{11}b_{22} + b_{12}b_{21})h_{12,t-1} + b_{21}b_{22}h_{22,t-1} \tag{7}$$

$$h_{22,t} = c_{21}^2 + c_{22}^2 + a_{12}^2 \varepsilon_{1,t-1}^2 + 2a_{12}a_{22}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{22}^2 \varepsilon_{2,t-1}^2 + b_{12}^2 h_{11,t-1} + 2b_{12}b_{22}h_{12,t-1} + b_{22}^2 h_{22,t-1} \tag{8}$$

Interpretation

- Cross terms like $a_{21}^2 \varepsilon_{2,t-1}^2$ in $h_{11,t}$: shock in asset 2 affects asset 1's variance = **volatility spillover**.

Diagonal and Scalar BEKK

Diagonal BEKK

- Restrict A and B to be **diagonal**:

$$\mathbf{H}_t = \mathbf{C}\mathbf{C}' + \mathrm{diag}(\mathbf{a})'\boldsymbol{\varepsilon}_{t-1}\boldsymbol{\varepsilon}_{t-1}'\mathrm{diag}(\mathbf{a}) + \mathrm{diag}(\mathbf{b})'\mathbf{H}_{t-1}\mathrm{diag}(\mathbf{b})$$

- Parameters: $N(N+1)/2 + 2N$. Element-wise: $h_{ij,t} = c_{ij}^* + a_i a_j \varepsilon_{i,t-1}\varepsilon_{j,t-1} + b_i b_j h_{ij,t-1}$

Scalar BEKK

- Further restrict: $\mathbf{A} = a\mathbf{I}_N$, $\mathbf{B} = b\mathbf{I}_N$:

$$\mathbf{H}_t = \mathbf{C}\mathbf{C}' + a^2\boldsymbol{\varepsilon}_{t-1}\boldsymbol{\varepsilon}_{t-1}' + b^2\mathbf{H}_{t-1}$$

- Parameters: $N(N+1)/2 + 2$. All pairs share the **same dynamics**.

Intuition: "one knob for shock sensitivity, one for memory" — identical across all pairs.

Trade-off

- Diagonal/Scalar BEKK: feasible for large N, but **no cross-asset spillovers**.

The Kronecker Product

Definition

- Let $\mathrm{A} \in \mathbb{R}^{m \times n}$ and $\mathrm{B} \in \mathbb{R}^{p \times q}$. The **Kronecker product** $\mathrm{A} \otimes \mathrm{B}$ is the $mp \times nq$ block matrix:

$$\mathrm{A} \otimes \mathrm{B} = \begin{pmatrix} a_{11}\mathrm{B} & \cdots & a_{1n}\mathrm{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathrm{B} & \cdots & a_{mn}\mathrm{B} \end{pmatrix}$$

- Dimensions do **not** need to match (unlike Hadamard)

Example: 2×2

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix} \quad \text{(result is } 4 \times 4\text{)}$$

Role in Multivariate GARCH

- BEKK stationarity condition: $\rho(\mathrm{A} \otimes \mathrm{A} + \mathrm{B} \otimes \mathrm{B}) < 1$
- Unconditional covariance: $\mathrm{vech}(\bar{\mathrm{H}}) = (\mathrm{I} - \mathrm{A} \otimes \mathrm{A} - \mathrm{B} \otimes \mathrm{B})^{-1}\mathrm{vech}(\mathrm{CC}')$

BEKK: Stationarity and Unconditional Covariance

Stationarity Condition

- The BEKK(1,1) process is covariance stationary if and only if:

$$\rho(\mathrm{A} \otimes \mathrm{A} + \mathrm{B} \otimes \mathrm{B}) < 1$$

- where $\rho(\cdot)$ denotes the spectral radius and $\otimes$ is the Kronecker product

Intuition: the multivariate analogue of $\alpha + \beta < 1$ — shocks must decay, not explode.

Unconditional Covariance Matrix

- If the stationarity condition holds:

$$\mathrm{vech}(\bar{\mathrm{H}}) = (\mathrm{I}_{N^*} - \mathrm{A} \otimes \mathrm{A} - \mathrm{B} \otimes \mathrm{B})^{-1} \mathrm{vech}(\mathrm{CC}')$$

Intuition: the long-run covariance matrix that H_t gravitates toward — like $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ in univariate GARCH.

BEKK: Strengths, Weaknesses, and Variant Selection

Strengths

- **Positive definiteness by construction** — no additional restrictions needed
- Captures **volatility spillovers** — the only class that directly models them
- Direct interpretation of A and B: shock transmission and persistence

Weaknesses

- **Parameter explosion**: full BEKK for $N = 10 \Rightarrow 255$ parameters
- Numerically **unstable** optimization — multiple local optima, slow convergence
- Results can be **sensitive to starting values**
- Cannot be decomposed into stages — joint estimation required

Practical Rule

- $N \leq 3$: full BEKK (if spillovers matter)
- $3 < N \leq 10$: diagonal BEKK
- $N > 10$: switch to DCC or factor models

BEKK: Key Insight and Common Pitfalls

: BEKK is a "smart reparametrization" of VEC

- BEKK does not add new economics — it writes the *same* dynamics in a p.d.-guaranteed form
- A′MA is p.s.d. by linear algebra, not economic theory; trades flexibility for convenience

: using full BEKK for $N > 5$

- Full BEKK for $N = 10$: **255 parameters** — optimizer finds local optima, results are fragile
- Caporin & McAleer (2012): "full BEKK is essentially uninterpretable for $N > 3$"
- Diagonal BEKK is often the practical choice, but then spillovers vanish — defeating BEKK's purpose

- $N > 5$: use DCC or factor models instead
- When you don't care about *direction* of spillovers (DCC captures co-movement without spillover channels)
- When you need fast re-estimation (BEKK: hours vs. DCC: seconds for $N = 50$)

BEKK Section Summary

3 Things to Remember

1. BEKK guarantees p.d. via the quadratic form A′MA — the **only** model that directly captures spillovers
2. Full BEKK: rich but infeasible for $N > 5$; diagonal BEKK: scalable but no spillovers
3. Estimation requires joint optimization — no two-step shortcut

Model Positioning

- **Use when**: $N \leq 5$ and volatility spillover *direction* matters
- **vs. VEC**: same flexibility, guaranteed p.d. **vs. DCC**: DCC scales better but cannot identify spillover direction

Key takeaway: *Use BEKK when* $N \leq 5$ *and you need to know* ***who transmits volatility to whom****.*

From BEKK to CCC: The Decomposition That Changed Everything

What BEKK taught us — and where it breaks down

- BEKK **solved** the PD problem and models volatility spillovers directly
- But: $N(N+1)/2 + 2N^2$ parameters $\Rightarrow$ estimation collapses for $N > 5$
- Joint estimation of all parameters — no decomposition into simpler sub-problems
- Practitioners abandoned BEKK for large portfolios by the late 1990s

Bollerslev's Insight (1990): Separate What Changes from What Doesn't

- **Observation**: variances change every day, but *correlations* move slowly
- **Idea**: decompose $H_t = D_t R D_t$ where D_t varies but R is fixed
- This separates a **hard** joint problem into N **easy** univariate problems + one correlation estimate

The Trade-off

- Gained: scalability ($3N + N(N-1)/2$ params), two-step estimation, handles $N > 100$
- Lost: volatility spillovers, time-varying correlations (constant R!)
- This trade-off motivated CCC — and later DCC to restore dynamic correlations

CCC: The Decomposition $H_t = D_t R D_t$

One sentence: *Covariances change only because volatilities change — the "wiring" between assets stays fixed.*

Key Idea: Separate Variances from Correlations

- Decompose the conditional covariance matrix:

$$H_t = \underbrace{D_t}_{\text{volatilities}} \underbrace{R}_{\text{correlations}} \underbrace{D_t}_{\text{volatilities}}$$

- $D_t = \text{diag}(\sqrt{h_{11,t}}, \ldots, \sqrt{h_{NN,t}})$ — conditional standard deviations (time-varying)
- R — **constant** correlation matrix (time-invariant)
- Each $h_{ii,t}$ follows a **univariate GARCH(1,1)**: $h_{ii,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i h_{ii,t-1}$

CCC: Properties and Parameter Count

One sentence: *CCC reduces a massive joint problem to N simple univariate GARCHs plus one correlation matrix.*

Properties

- H_t is p.d. iff R is p.d. and all $h_{ii,t} > 0$
- Covariances: $h_{ij,t} = \rho_{ij}\sqrt{h_{ii,t}}\sqrt{h_{jj,t}}$
- Parameters: $3N$ (univariate GARCH) $+N(N-1)/2$ (correlations)

Comparison

- $N = 50$: CCC needs **1,375** parameters vs. BEKK's $\approx 6{,}275$
- Two-step estimation: each step is low-dimensional

Key takeaway: *CCC assumes correlations are fixed — covariances change only because volatilities change.*

CCC: Two-Step Estimation

Step 1: Estimate Univariate GARCH Models

- For each asset $i = 1, \ldots, N$:

1. Estimate a GARCH(1,1) model for r_{it}
2. Obtain the standardized residuals: $z_{it} = \varepsilon_{it} / \sqrt{h_{ii,t}}$

Step 2: Estimate the Correlation Matrix

$$\hat{\mathrm{R}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{z}_t \mathbf{z}_t'$$

- where $\mathbf{z}_t = (z_{1t}, \ldots, z_{Nt})'$ are the standardized residuals from Step 1

Advantage

- Each step involves **low-dimensional optimization**
- Step 1: N univariate optimizations (3 params each); Step 2: sample correlation — no optimization

CCC: Strengths and Limitations

Strengths

- **Parsimony**: few parameters, easy to estimate even for large N
- **Positive definiteness**: guaranteed by construction
- **Interpretability**: time-varying covariances driven solely by changing variances
- Good **baseline model** for comparison

Limitations

- **Constant correlations** are empirically rejected in most financial data
- Cannot capture **contagion** or **flight-to-quality** effects
- Covariances change only because variances change, not because the **relationship** between assets changes
- Tse (2000) and Engle (2002) proposed tests for constant correlation

Key takeaway: *CCC is the ideal benchmark — simple, fast, and always your starting point.*

CCC: Key Insight and Common Pitfalls

: CCC made MGARCH practical — not accurate

- CCC's genius is **computational**: fixing R decomposes the problem into N univariate GARCHs
- CCC is rarely the "right" model, but always the right **benchmark**
- If DCC cannot beat CCC out-of-sample, the extra complexity is wasted

: using CCC for risk management during crises

- CCC **underestimates** portfolio VaR during stress: true correlations spike, but CCC uses the sample average
- Empirically: CCC underestimates crisis VaR by 20–30% vs. DCC (see Case Study 1)
- Regulators (Basel III/IV) expect models to capture **time-varying** dependence

:

- Short horizons (< 1 year) where correlations haven't changed much
- Asset pairs with genuinely stable correlations (e.g., same-sector stocks)
- As a **diagnostic tool**: if CCC and DCC give similar results, correlations are stable

CCC Section Summary

3 Things to Remember

1. CCC decomposes $H_t = D_t R D_t$ — **variances move, correlations don't**
2. Two-step estimation: N univariate GARCHs + sample correlation — fast, scalable, guaranteed p.d.
3. Constant correlations are empirically **rejected** in most financial data

Model Positioning

- **Use as**: the default **benchmark** — if DCC can't beat it, complexity is wasted
- **vs. BEKK**: much simpler, no spillovers **vs. DCC**: same structure, but R is fixed

Key takeaway: *Always estimate CCC first — it is the null hypothesis that DCC must beat.*

Evidence Against Constant Correlations

One sentence: *CCC's constant-correlation assumption is rejected in virtually all financial data.*

The empirical evidence

- CCC assumes ρ_{ij} is **fixed** — but crises repeatedly violate this
- S&P 500–DAX correlation: ≈ 0.40 in calm markets, > 0.85 during GFC (2008)
- Stock–bond correlation **flipped sign**: $+0.30$ pre-2000 $\rightarrow$ -0.30 post-2000 (flight to quality)

Statistical tests

- Tse (2000) and Engle–Sheppard (2001): constant correlation rejected at 1% in most pairs
- CCC underestimates portfolio VaR during stress by 20–30% vs. DCC

Engle's Solution (2002): Make R_t Dynamic

One sentence: *Keep CCC's structure, but let correlations respond to shocks — for just 2 extra parameters.*

The DCC idea

- Replace R with R_t — correlations now respond to joint shocks
- Key trick: model a proxy Q_t (which is *not* a correlation matrix), then rescale
- **Only 2 extra parameters** (a, b) regardless of portfolio size N

What DCC preserves from CCC

- Same two-step estimation: Step 1 is identical (univariate GARCHs)
- Same scalability: $3N + N(N-1)/2 + 2$ parameters — just 2 more than CCC
- If $a = b = 0$, DCC **reduces exactly to CCC** — CCC is nested

DCC in One Sentence

DCC = CCC with dynamic correlations

$H_t = D_t R_t D_t$ where R_t follows a GARCH-like process with **2 parameters** (a, b)

	CCC	**DCC**	**Difference**
Correlations	constant R	dynamic R_t	+2 parameters
Estimation	2-step	2-step	same structure
$N = 50$ params	1,375	**1,377**	negligible

The DCC Model (Engle, 2002)

: DCC = CCC + dynamic correlations + only 2 extra parameters

- Same decomposition: $H_t = D_t R_t D_t$
- Same D_t (univariate GARCH), but R_t is now **time-varying**

Correlation Dynamics (the only new equation)

- R_t is derived from a pseudo-correlation matrix Q_t:

$$Q_t = (1 - a - b)\bar{Q} + a\, z_{t-1} z'_{t-1} + b\, Q_{t-1} \tag{9}$$

$$R_t = \text{diag}(Q_t)^{-1/2}\, Q_t\, \text{diag}(Q_t)^{-1/2} \tag{10}$$

- $\bar{Q}$: unconditional correlation; a: news impact; b: persistence; $a + b < 1$

Intuition: correlations spike when assets move together and mean-revert to $\bar{Q}$ otherwise.

DCC: Intuition and Interpretation

The Q_t Process

$$Q_t = \underbrace{(1-a-b)\bar{Q}}_{\text{mean reversion}} + \underbrace{a\, z_{t-1} z_{t-1}'}_{\text{news impact}} + \underbrace{b\, Q_{t-1}}_{\text{persistence}}$$

- a measures how **quickly correlations react** to new shocks
- b measures the **persistence** of correlations
- $a + b$ close to 1: correlations are highly persistent (close to IGARCH-like behavior)
- If $a = b = 0$: DCC reduces to CCC

The Rescaling Step

- Q_t is **not** a proper correlation matrix (diagonal $\neq 1$)
- The rescaling $R_t = \text{diag}(Q_t)^{-1/2} Q_t \,\text{diag}(Q_t)^{-1/2}$ ensures $R_{ii,t} = 1$
- This guarantees $-1 \leq R_{ij,t} \leq 1$ and R_t is p.d. (if Q_t is p.d.)

DCC: Two-Step Estimation

Step 1: Univariate GARCH (Same as CCC)

1. Estimate GARCH(1,1) for each asset i separately
2. Compute standardized residuals $z_{it} = \varepsilon_{it} / \sqrt{h_{ii,t}}$
3. Compute $\bar{\mathbf{Q}} = T^{-1} \sum_t \mathbf{z}_t \mathbf{z}_t'$

Step 2: Estimate DCC Parameters (a, b)

- Maximize the **correlation component** of the log-likelihood:

$$\mathcal{L}_{\mathbf{2}}(a, b) = -\frac{1}{2} \sum_{t=\mathbf{1}}^{T} \left[\log |\mathbf{R}_t| + \mathbf{z}_t' \mathbf{R}_t^{-\mathbf{1}} \mathbf{z}_t - \mathbf{z}_t' \mathbf{z}_t \right]$$

- A **2-parameter** optimization regardless of N!

Intuition: only the "correlation part" of the likelihood — volatility was handled in Step 1.

Composite Likelihood

- Consistent but not fully efficient. SEs require Bollerslev–Wooldridge sandwich estimator.

DCC: Bivariate Example

Example: Two Assets ($N = 2$)

- With $\mathbf{z}_t = (z_{1t}, z_{2t})'$, the DCC dynamics simplify to:

$$q_{12,t} = (1 - a - b)\bar{q}_{12} + a\, z_{1,t-1} z_{2,t-1} + b\, q_{12,t-1} \tag{11}$$

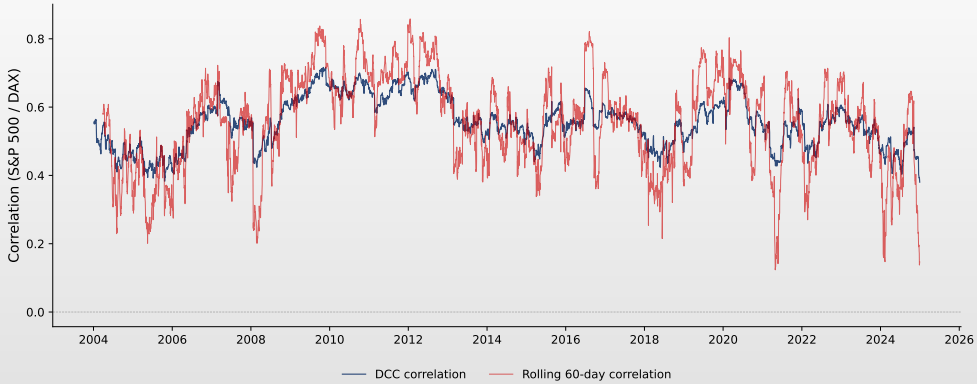
$$\rho_{12,t} = \frac{q_{12,t}}{\sqrt{q_{11,t}\, q_{22,t}}} \tag{12}$$

- where $q_{11,t} = (1 - a - b) + a\, z_{1,t-1}^2 + b\, q_{11,t-1}$ (and similarly for $q_{22,t}$)

Interpretation

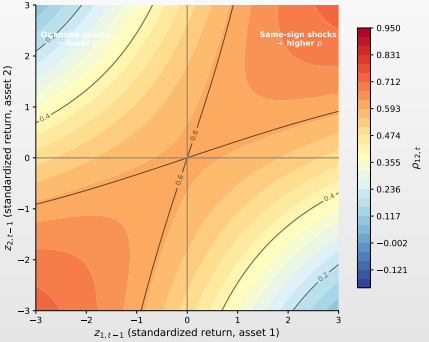
- If both $z_{1,t-1}$ and $z_{2,t-1}$ are large and have the **same sign**: $q_{12,t}$ increases $\Rightarrow$ correlation increases
- Joint negative shocks (crises) **push correlations upward** — captures contagion
- Correlations **mean-revert** to $\bar{q}_{12}$ at rate $(1 - a - b)$

Case Study: DCC vs. Rolling Correlation (S&P 500 – DAX)



TSA_ch5b_dcc_estimation

News Impact on DCC Correlation

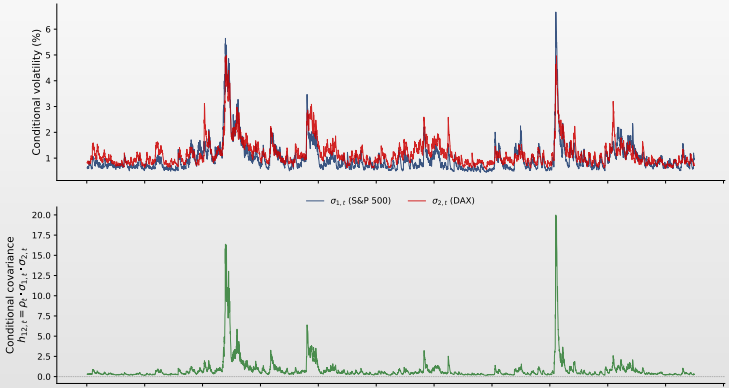


Interpretation

- When both assets experience shocks of the **same sign**, correlation **increases**
- **Opposite-sign** shocks reduce correlation — decoupling effect

TSA_ch5b_news_impact_corr

Covariance Decomposition: $H_t = D_t R_t D_t$



TSA_ch5b_cov_decomp

Numerical Example: DCC Step by Step

Data: S&P 500 and DAX, day $t-1$

- Standardized residuals: $z_{1,t-1} = -2.1$ (S&P 500), $z_{2,t-1} = -1.8$ (DAX)
- Parameters: $a = 0.05$, $b = 0.93$, $\bar{q}_{12} = 0.65$
- Previous state: $q_{11,t-1} = 1.02$, $q_{22,t-1} = 0.98$, $q_{12,t-1} = 0.68$

Step 1: Update Q_t

$$q_{11,t} = (1 - 0.05 - 0.93) \cdot 1 + 0.05 \cdot (-2.1)^2 + 0.93 \cdot 1.02 = 0.02 + 0.2205 + 0.9486 = 1.189$$

$$q_{22,t} = 0.02 + 0.05 \cdot (-1.8)^2 + 0.93 \cdot 0.98 = 0.02 + 0.162 + 0.9114 = 1.093$$

$$q_{12,t} = 0.02 \cdot 0.65 + 0.05 \cdot (-2.1)(-1.8) + 0.93 \cdot 0.68 = 0.013 + 0.189 + 0.6324 = 0.834$$

Step 2: Rescale $\rightarrow$ correlation $\rho_{12,t}$

$$\rho_{12,t} = \frac{q_{12,t}}{\sqrt{q_{11,t} \cdot q_{22,t}}} = \frac{0.834}{\sqrt{1.189 \cdot 1.093}} = \frac{0.834}{1.140} = 0.732$$

Correlation rose from ≈ 0.67 to 0.73 due to joint negative shocks!

DCC vs CCC: When Does It Matter?

Testing for Constant Correlation

- **Engle–Sheppard (2001) test**: H_0: correlations are constant
  - Regress $\text{vech}(z_t z_t' - \hat{R})$ on lagged values
  - $T \cdot R^2 \sim \chi^2$ under H_0

Empirical Evidence

- CCC is almost **always rejected** in financial data
- DCC provides better out-of-sample portfolio performance
- The improvement is largest during **crisis periods**

Practical Rule

- Use CCC as a **benchmark** and DCC as the **workhorse** model. If the Engle–Sheppard test rejects CCC, use DCC.

The Tse (2000) LM Test for Constant Correlation

Hypotheses

- H_0: $\rho_{ij,t} = \rho_{ij}\ \forall t$ (correlations are **constant** — CCC model is correct)
- H_1: $\rho_{ij,t}$ varies over time (a DCC model is needed)

Test Construction

- Estimate CCC and obtain standardized residuals $\hat{z}_{it} = \hat{\varepsilon}_{it}/\hat{\sigma}_{it}$
- Compute pairwise products: $\hat{u}_{ij,t} = \hat{z}_{it}\hat{z}_{jt} - \hat{\rho}_{ij}$ for each pair (i, j)
- Regress $\hat{u}_{ij,t}$ on p lagged values $\hat{u}_{ij,t-1}, \ldots, \hat{u}_{ij,t-p}$
- Test statistic: $LM = T \sum_{i<j} R_{ij}^2 \xrightarrow{d} \chi^2\left(\frac{N(N-1)}{2} \cdot p\right)$ under H_0

Difference from Engle–Sheppard (2001)

- **Engle–Sheppard**: regresses the vector $\text{vech}(z_t z_t' - \hat{R})$ **jointly** on lagged values — tests all pairs in a single regression
- **Tse**: tests each pair (i, j) **separately**, then combines the statistics — easier to interpret but less powerful for complex dependencies

DCC: Key Insight and Common Pitfalls

: DCC captures correlation dynamics with just 2 parameters

- a and b are **shared across all pairs** — for 100 assets, 4,950 correlations share 2 parameters
- This is simultaneously its greatest strength (scalability) and limitation (homogeneity)

- **Ignoring Aielli (2013)**: original DCC is inconsistent ($E[Q_t] \neq \bar{Q}$). Use cDCC.
- **Same-speed assumption**: stock–bond fast, stock–stock slow — single (a, b) compromises
- **Forgetting sandwich**: naive SEs from two-step are **wrong** — use Bollerslev–Wooldridge

- Need **spillover direction**: use BEKK or Diebold–Yilmaz instead
- Correlations **genuinely constant**: DCC adds noise; CCC is more efficient
- Assets have very **different correlation dynamics**: consider block-DCC or copula

DCC Section Summary

3 Things to Remember

1. DCC = CCC with dynamic R_t: only **2 extra parameters** (a, b) for all $N(N-1)/2$ correlations
2. Q_t equation is a GARCH(1,1) for correlations: a = shock reactivity, b = persistence
3. Two-step estimation is consistent but needs **sandwich standard errors**

Model Positioning

- **Use as**: the **industry workhorse** — best balance of flexibility and scalability
- **vs. CCC**: captures crisis correlation spikes **vs. BEKK**: no spillover direction, but scales to $N > 100$

Key takeaway: *DCC is the industry standard — 2 parameters capture all $N(N-1)/2$ dynamic correlations.*

Asymmetric DCC (aDCC) – Cappiello, Engle & Sheppard (2006)

Motivation: Leverage in Correlations

- Correlations may respond asymmetrically to negative vs. positive shocks ("correlation leverage"). The aDCC extends DCC:

$$Q_t = (\bar{Q} - a\bar{Q} - b\bar{Q} - g\bar{N}) + a\, z_{t-1} z_{t-1}' + b\, Q_{t-1} + g \underbrace{n_{t-1} n_{t-1}'}_{\text{asymmetric term}}$$

- where $n_t = z_t \odot 1(z_t < 0)$ and $\bar{N} = T^{-1} \sum_t n_t n_t'$

Intuition: joint losses (g term) can increase correlations more than joint gains — the "bad news" amplifier.

Properties

- Adds only **one parameter** (g) to the DCC model
- Captures the empirical finding that correlations increase more after **joint negative shocks**
- Nests DCC ($g = 0$) and CCC ($a = b = g = 0$) as special cases

GO-GARCH (van der Weide, 2002)

Idea: Generalized Orthogonal GARCH

- Map the N-dimensional return vector to N **uncorrelated factors**:

$$\boldsymbol{\varepsilon}_t = \mathbf{Z}\mathbf{f}_t, \qquad \mathbf{f}_t = (f_{1t}, \ldots, f_{Nt})'$$

- where $\mathbf{Z}$ is a mixing matrix and the factors f_{it} are **uncorrelated** with individual GARCH dynamics:

$$\sigma^2_{f_i,t} = \omega_i + \alpha_i f^2_{i,t-1} + \beta_i \sigma^2_{f_i,t-1}$$

Reconstruction

$$\mathbf{H}_t = \mathbf{Z}\,\mathrm{diag}(\sigma^2_{f_1,t}, \ldots, \sigma^2_{f_N,t})\,\mathbf{Z}'$$

Intuition: model volatility in "factor space" (simple), then rotate back to "asset space" (complex).

- Positive definite **by construction**
- Parameters: $N(N-1)/2 + 3N$ — scales well with N
- Related to Independent Component Analysis (ICA)

Factor GARCH Models

Idea: Reduce Dimensionality with Factors ($K \ll N$)

$$\boldsymbol{\varepsilon}_t = \tilde{\mathbf{f}}_t + \mathbf{u}_t$$

- $\tilde{}$: $N \times K$ loadings; $\mathbf{f}_t$: $K \times 1$ factors; $\mathbf{u}_t$: idiosyncratic

$$\mathbf{H}_t = \underbrace{\tilde{\mathbf{H}}_t^f \tilde{}'}_{\text{systematic risk}} + \underbrace{\boldsymbol{\Sigma}_t^u}_{\text{idiosyncratic risk}}$$

Intuition: model a few common factors, treat the rest as noise.

Variants

- **O-GARCH** (Alexander, 2001): PCA + diagonal GARCH on principal components
- **Factor GARCH** (Engle, Ng & Rothschild, 1990): parametric factor structure
- **DCC-MIDAS** (Colacito, Engle & Ghysels, 2011): mixed-frequency correlations

When to Use

- The **go-to approach** for large N (50–500 assets): parsimony + economic interpretability.

Tse–Tsui DCC (2002)

An Alternative to Engle's DCC

- Tse & Tsui (2002) propose a different dynamics for R_t:

$$R_t = (1 - \theta_1 - \theta_2) \underbrace{R}_{\text{long-run}} + \theta_1 \underbrace{\Psi_{t-1}}_{\text{local window}} + \theta_2 \underbrace{R_{t-1}}_{\text{persistence}}$$

- *Intuition: blend long-run, recent-window, and yesterday's correlation — the result is automatically a valid correlation matrix.*
- Ψ_{t-1} = sample correlation over the last M observations

Comparison with Engle's DCC

- **Advantage**: R_t is a proper correlation matrix **by construction** (no rescaling needed)
- **Disadvantage**: requires choosing window M; less used in practice
- Both models yield similar empirical results

Copula-GARCH: A Flexible Approach

Idea: Separate Margins from Dependence Structure

- **Sklar's Theorem**: $F(x_1, \ldots, x_N) = C\big(F_1(x_1), \ldots, F_N(x_N)\big)$
- *Intuition: model each asset's risk profile separately, then "glue" them together with a copula.*
- F_i = marginal distributions (GARCH + flexible dist.); C = copula (dependence structure)

Advantages over DCC

- Allows **asymmetric tail dependence**; margins can have **different** distributions (Student-t, skewed-t, GED)
- t-Student copula: DCC-like ρ_t + tail dependence parameterized by ν

Implementation

- R: `copula` + `rugarch`; Python: `copulae`

EWMA and RiskMetrics

Exponentially Weighted Moving Average (EWMA)

- The simplest multivariate volatility model — no estimation needed:

$$H_t = (1-\lambda)\boldsymbol{\varepsilon_{t-1}\varepsilon'_{t-1}} + \lambda H_{t-1}, \qquad \lambda = 0.94 \text{ (daily)}$$

Intuition: weighted average of yesterday's forecast and realized shocks — no estimation required.

Connection to MGARCH

- EWMA $\equiv$ Scalar BEKK with $\omega = 0$, $a^2 = 1-\lambda$, $b^2 = \lambda$
- Also equivalent to IGARCH ($\alpha + \beta = 1$) without intercept
- RiskMetrics (J.P. Morgan, 1996) used this as the **industry standard**

Limitations

- No estimation $\Rightarrow$ cannot adapt to different market conditions
- $\lambda = 0.94$ may not be optimal for all asset classes
- No asymmetric effects, no mean reversion in variance

Model Taxonomy: A Summary

Model	P.D.	Spillovers	Dynamic ρ	Max N
VEC	No*	Yes	Yes	~ 3
Diagonal VEC	Yes*	No	Yes	~ 10
BEKK (full)	Yes	Yes	Yes	~ 5
Diagonal BEKK	Yes	No	Yes	~ 20
Scalar BEKK / EWMA	Yes	No	Yes	~ 100
CCC	Yes	No	No	~ 100
DCC	Yes	No	Yes	~ 500
aDCC	Yes	No	Yes	~ 500
GO-GARCH	Yes	No	Yes	~ 50
Factor GARCH	Yes	No	Yes	~ 500

* Requires additional parameter restrictions.

- The choice of model depends on: (1) number of assets, (2) need for spillover effects, (3) need for dynamic correlations, (4) computational budget.

Multivariate GARCH Model Hierarchy

Nesting Relationships

- EWMA $\subset$ Scalar BEKK $\subset$ Diagonal BEKK $\subset$ Full BEKK $\subset$ Full VEC
- CCC $\subset$ DCC $\subset$ aDCC
- DCC and Diagonal BEKK: **not** nested in each other

Choosing by Objective

Objective	Recommended Model	Why
Portfolio VaR	DCC / aDCC	Dynamic correlations, scalable
Optimal hedging	DCC / Diagonal BEKK	Time-varying covariance
Shock transmission	Full BEKK	Captures spillovers
$N > 100$ assets	Factor GARCH / DCC	Parsimony, feasibility
Correlation asymmetry	aDCC	Leverage term g
Tail dependence	Copula-GARCH	Distributional flexibility

When MGARCH Models Fail

Model Misspecification

- All MGARCH assume a **parametric structure** for H_t — the true DGP may not belong to this class
- DCC: **one correlation dynamic for all pairs** (a, b scalars), yet equity-bond and equity-equity behave differently
- BEKK: **quadratic** dynamics — cannot capture regime switches or structural breaks
- Gaussian QML: **consistent but inefficient** under fat tails — losses grow with N

Tail Risk Failures

- Normal innovations **underestimate joint tail risk**: Pr(all assets crash) too low
- Student-t copula: **symmetric** tail dependence — assets crash together but don't boom together
- VaR violations in crises are **clustered** — even DCC with correct coverage has concentrated violations

Key Insight

- MGARCH: **local approximations** — work well in stable regimes, underperform during regime transitions

Crisis Instability and the DCC Paradox

The DCC Paradox

- DCC is **most needed** during crises — but crises are precisely when it is **least reliable**
- Parameter estimates from tranquil periods may **not apply** to crisis dynamics
- Aielli (2013): standard DCC estimator has **bias** in two-step procedure — cDCC corrects this

Structural Breaks in Correlations

- **Euro (1999)**: European equity correlations shifted permanently upward
- **GFC (2008)**: cross-asset correlations jumped and did not fully revert for years
- **COVID (2020)**: unprecedented correlation spike followed by rapid decorrelation
- DCC mean-reversion ($a + b < 1$) forces correlations to $\bar{Q}$ — can be **wrong** after structural breaks

Remedies

- **DCC-MIDAS** (Colacito, Engle & Ghysels, 2011): separates short-run and long-run correlation
- **Markov-Switching DCC**: discrete regime changes in correlation dynamics
- **Rolling windows**: re-estimate every 1–2 years to capture structural shifts

Overfitting vs. Parsimony: The Bias-Variance Trade-off

The Overfitting Trap

- Full BEKK for $N = 5$: **55 parameters** — fits in-sample perfectly, poor out-of-sample
- aDCC asymmetry ($+1$ parameter) helps for equities but may overfit bond-commodity pairs
- Student-t d.f. ν can **compensate** for correlation misspecification, masking the real problem

Empirical Evidence (Caporin & McAleer, 2012; Laurent et al., 2012)

- **Scalar BEKK** $\approx$ **DCC** in out-of-sample forecasting for most portfolios
- Adding complexity beyond DCC(1,1) rarely improves portfolio VaR or hedging effectiveness
- The **simple EWMA** ($\lambda = 0.94$) remains a competitive benchmark — hard to beat consistently
- Information criteria (AIC/BIC) favor DCC over BEKK for $N > 3$

Practical Rule

- Start simple (EWMA or CCC), add complexity **only if out-of-sample tests justify it**
- Use **Model Confidence Sets** (Hansen, Lunde & Nason, 2011) rather than single-model selection

Empirical Model Comparison: What the Literature Shows

Out-of-Sample Forecasting Performance (Based on Laurent et al., 2012; Caporin & McAleer, 2014)

Model	Portfolio VaR	Hedge	GMVP	Speed	Max N
EWMA ($\lambda = 0.94$)	Good	Fair	Fair	⋆ ⋆ ⋆ ⋆ ⋆	1000+
CCC-GARCH	Fair	Fair	Poor	⋆ ⋆ ⋆⋆	500
DCC-GARCH	**Good**	**Good**	Good	⋆ ⋆ ⋆⋆	500
aDCC-GARCH	**Good**	**Good**	**Good**	⋆ ⋆ ⋆	300
Diagonal BEKK	Good	Good	Good	⋆ ⋆ ⋆	20
Full BEKK	Fair*	Good	Fair*	⋆	5
GO-GARCH	Good	Fair	Good	⋆⋆	50
Copula-DCC	**Best**	Good	Good	⋆⋆	100

- *Full BEKK overfits in-sample, degrades out-of-sample for $N > 3$. GMVP = Global Minimum Variance Portfolio.

Key Findings

- DCC is the **best risk-adjusted choice** across most applications
- Copula-DCC dominates for **tail risk** (ES, extreme VaR) but at higher computational cost
- Complexity beyond aDCC yields **diminishing returns** for most portfolios

Estimation Methods

Maximum Likelihood Estimation (MLE)

- **Full MLE**: jointly estimate all parameters — optimal but computationally expensive
- **Two-step (Composite) MLE**: estimate univariate models first, then correlation parameters
  - Used by CCC, DCC, aDCC
  - Consistent but less efficient than full MLE

Quasi-Maximum Likelihood (QML)

- Use the Gaussian likelihood even if the true distribution is non-Gaussian
- QML estimator is **consistent** under mild regularity conditions
- Use **robust (sandwich) standard errors**: $\hat{V} = \hat{J}^{-1}\hat{I}\hat{J}^{-1}$

Alternative: Multivariate Student-t

- Better captures fat tails: $z_t \sim t_\nu(0, I_N)$
- Adds one parameter (ν) for degrees of freedom

The Sandwich Estimator

Why Is It Needed?

- QML assumes Gaussian distribution, but financial data exhibit **fat tails** and **asymmetry**
- Parameters are **consistent**, but classical standard errors are **incorrect**
- Solution: the **sandwich** estimator of Bollerslev–Wooldridge (1992) corrects the standard errors

The "Sandwich" Structure — Where the Name Comes From

- Variance-covariance matrix of the estimator: $\hat{V}(\hat{\boldsymbol{\theta}}) = \underbrace{\hat{\mathsf{J}}^{-1}}_{\text{"bread"}} \underbrace{\hat{\mathsf{I}}}_{\text{"filling"}} \underbrace{\hat{\mathsf{J}}^{-1}}_{\text{"bread"}}$
- $\hat{\mathsf{J}} = -\frac{1}{T}\sum_t \frac{\partial^2 \ell_t}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}$ (the Hessian — assumes the model is correct)
- $\hat{\mathsf{I}} = \frac{1}{T}\sum_t \frac{\partial \ell_t}{\partial \boldsymbol{\theta}} \frac{\partial \ell_t}{\partial \boldsymbol{\theta}'}$ (outer product of gradients — *empirical*, no assumptions)
- *Intuition:* $\hat{\mathsf{J}}$ *trusts the model;* $\hat{\mathsf{I}}$ *trusts the data. If correctly specified,* $\mathsf{I} = \mathsf{J}$ *and* $\hat{V} = \mathsf{J}^{-1}$.

When to Use

- **Always** under QML estimation (GARCH with Gaussian likelihood on non-Gaussian data)
- In two-step DCC estimation: stage 2 standard errors must be corrected for stage 1 uncertainty

Diagnostic Checks

Residual Diagnostics

- After estimation, compute standardized residuals $\hat{z}_t = \hat{D}_t^{-1}\hat{\varepsilon}_t$ and check:

1. **No serial correlation** in $\hat{z}_{it}$ and $\hat{z}_{it}^2$ (Ljung–Box test for each series)
2. **No cross-correlation** in $\hat{z}_{it}\hat{z}_{jt}$ (Hosking, 1980; Li & McLeod, 1981)
3. **Normality** of $\hat{z}_t$ (if Gaussian likelihood assumed)

Model Comparison

- **Information criteria**: AIC, BIC (*Akaike/Bayesian Information Criterion*) (penalize over-parameterization)
- **Likelihood ratio tests**: nested models (CCC vs DCC)
- **Out-of-sample evaluation**: portfolio performance, VaR backtesting
- **Mincer–Zarnowitz regression**: forecast $\hat{h}_{ij,t}$ vs realized covariance

Practical Estimation: Tips and Pitfalls

Practical Tips

- **Starting values**: initialize from OLS (*Ordinary Least Squares*) estimates or EWMA ($\lambda = 0.94$)
- **Returns**: use log-returns $\times 100$ (in percent) — stabilizes optimization
- **Missing data**: align series on common dates; do not interpolate
- **Convergence**: run optimization from multiple starting points

Common Pitfalls

- **Nonsynchronous trading**: DAX closes at 17:30 CET, NYSE at 16:00 EST — intraday correlations are underestimated
  - Solution: Scholes–Williams (1977) correction or open-to-open returns
- **Structural breaks**: crises permanently alter correlation regimes — DCC assumes a single regime
- **Student-t overfitting**: degrees of freedom ν may compensate for other specification issues

Model Selection: Practical Guidelines

Decision Flowchart

1. **How many assets?**
   - $N \leq 5$: BEKK, DCC, or aDCC
   - $5 < N \leq 50$: DCC, aDCC, GO-GARCH
   - $N > 50$: Factor GARCH, DCC-MIDAS, EWMA
2. **Do you need volatility spillovers?**
   - Yes: Full BEKK (small N) or Factor models
   - No: DCC family
3. **Is there asymmetry in correlations?**
   - Test for it $\Rightarrow$ if yes, use aDCC
4. **Validate**: check residual diagnostics, out-of-sample performance

Case Study 1: Volatility Spillovers Across Markets

Learning goal: how BEKK off-diagonals reveal the direction of shock transmission between markets.

Spillover Indicators

- From the BEKK model: $H_t = C'C + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + B'H_{t-1}B$
- **Spillover intensity** from asset j to asset i:

$$\text{Spillover}_{j\to i} = a_{ij}^2 + b_{ij}^2$$

- **Diebold–Yilmaz** index (2012): $S^H = \frac{1}{N}\sum_{i\neq j}\theta_{ij}^H$ where θ_{ij}^H = share of i's variance due to shock j

Methodology

- Data: daily returns S&P 500, DAX, FTSE 100, Nikkei 225 (2004–2024, $T \approx 5000$)
- Bivariate BEKK(1,1) for each pair of assets

Case Study 1: What is a Volatility Spillover?

Definition

- A **volatility spillover** occurs when a shock in market j increases the conditional variance of market i, beyond what i's own past shocks and variances predict

Transmission Channels

- **Direct financial linkages**: cross-border holdings, interbank lending, common counterparties
- **Information channel**: price discovery flows from developed to emerging markets
- **Behavioral channel**: herding, flight to quality, correlated margin calls
- **Time-zone effect**: NY close $\rightarrow$ Asian open $\rightarrow$ European open — sequential transmission

Why It Matters for Risk Management

- Portfolio VaR depends on **cross-market** covariance, not just individual volatilities
- During crises, all channels activate simultaneously $\Rightarrow$ correlation spike $\Rightarrow$ diversification fails
- Example: Lehman collapse (Sep 2008) $\rightarrow$ US vol triples $\rightarrow$ DAX opens 5% lower next day

Case Study 1: BEKK Parameter Matrix — Reading Off-Diagonals

The 2×2 BEKK Structure (bivariate case)

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

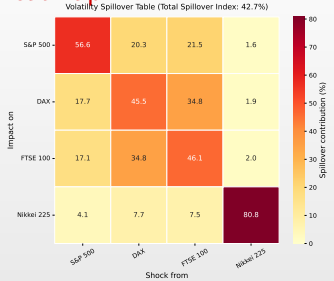
- a_{12}: shock in asset 2 $\Rightarrow$ **immediate** volatility increase in asset 1
- b_{12}: conditional variance of asset 2 $\Rightarrow$ **persistent** effect on variance of asset 1
- Total spillover: $\text{Spillover}_{2\to1} = a_{12}^2 + b_{12}^2$

Numerical Example: S&P 500 $\leftrightarrow$ DAX

	a_{12}	a_{21}	b_{12}	b_{21}
Estimate	0.18	0.09	0.31	0.15
Squared	0.032	0.008	0.096	0.023

- $\text{Spillover}_{\text{SP}\to\text{DAX}} = 0.008 + 0.023 = 0.031$ vs. $\text{Spillover}_{\text{DAX}\to\text{SP}} = 0.032 + 0.096 = 0.128$
- $\Rightarrow$ S&P $\to$ DAX is **4**$\times$ **stronger** than DAX $\to$ S&P

Case Study 1: Volatility Spillover Heatmap



Key Results

- S&P 500 is the main volatility **exporter** — US shocks propagate globally within 24h
- European markets (DAX, FTSE) are strongly **interconnected** (high reciprocal spillover)
- Nikkei is most **independent** — time zone lag reduces direct transmission
- **Implication**: a global portfolio cannot ignore the US → Europe channel

TSA_ch5b_spillover_heatmap

Case Study 1: Directed Spillovers — Who Exports, Who Imports?

Net Spillover = Transmitted − Received

Market	Transmitted	Received	Net	Role
S&P 500	66	55	$+11$	Net exporter
DAX	61	62	-1	Balanced
FTSE 100	58	60	-2	Slight importer
Nikkei	32	40	-8	Net importer

Interpretation

- S&P 500 is the **volatility hub**: it transmits more to every other market than it receives
- DAX and FTSE are nearly **balanced** — they are both recipients of US shocks and transmitters to each other
- Nikkei is most **isolated**: receives 8 p.p. more than it sends — time-zone lag acts as a natural barrier
- **Practical**: hedging US equity exposure is the single most impactful risk reduction action for a global portfolio

Case Study 1: Diebold–Yilmaz Spillover Table

VAR(5) on realized volatilities; horizon $H = 10$ days

From $\rightarrow$	S&P	DAX	FTSE	Nikkei	**Received**
S&P 500	45	22	20	13	55
DAX	25	38	28	9	62
FTSE 100	23	27	40	10	60
Nikkei	18	12	10	60	40
Transmitted	66	61	58	32	$S^H \approx 54\%$

Interpretation

- S&P 500: largest net **exporter** (transmitted $-$ received $= 66 - 55 = +11$ p.p.)
- DAX and FTSE: strong bilateral spillover (28%/27%) — tightly linked European markets
- Nikkei: most **isolated** ($32 - 40 = -8$ p.p.) — receives more than it transmits
- $S^H \approx 54\%$: over half of volatility variation is due to cross-market **contagion**

TSA_ch5b_spillover_heatmap

Case Study 1: Spillover Dynamics — Calm vs. Crisis

Rolling Diebold–Yilmaz Index ($H = 10$, 252-day window)

Period	S^H	Top channel	Event
2004–2006	42%	SP $\rightarrow$ DAX	Pre-crisis calm
2008–2009	72%	SP $\rightarrow$ all	Global Financial Crisis
2011–2012	65%	DAX $\rightarrow$ FTSE	European sovereign debt
2014–2019	48%	SP $\rightarrow$ DAX	Post-crisis recovery
2020 (Mar)	70%	SP $\rightarrow$ all	COVID-19 crash
2022	58%	SP $\rightarrow$ DAX	Monetary tightening

Key Insight

- Spillover index nearly **doubles** during crises (42% $\rightarrow$ 72%) — diversification collapses exactly when it is needed most
- The US market dominates spillover in **all** crisis episodes — no regime where Europe leads
- Recovery is asymmetric: post-GFC S^H took 3 years to normalize; post-COVID only 6 months

Case Study 1: Contagion Mechanisms During Crises

Why Does Spillover Increase in Crises?

- **Margin calls**: forced selling in one market $\Rightarrow$ deleveraging across all markets simultaneously
- **Liquidity spirals** (Brunnermeier & Pedersen, 2009): loss $\rightarrow$ margin call $\rightarrow$ liquidation $\rightarrow$ price drop $\rightarrow$ new margin call
- **Information contagion**: uncertainty about counterparty exposure creates *"who's next?"* dynamics
- **Flight to quality**: correlated exit from risky assets into safe havens (USD, Bunds, Gold)

Empirical Evidence from BEKK Estimates

- **GFC**: a_{12} (S&P$\rightarrow$DAX) increased from 0.12 to 0.31 — shock transmission *tripled*
- **COVID**: all off-diagonal b_{ij} spiked simultaneously — *persistent* cross-market volatility
- **Euro crisis**: asymmetric — DAX$\rightarrow$FTSE dominated, US relatively insulated

Case Study 1: Implications for Global Portfolio Construction

Practical Risk Management Lessons

- **Diversification is regime-dependent**: in calm periods, a 4-market portfolio reduces vol by 30%; in crises, only by 10%
- **US exposure dominates risk**: even a 25% US allocation drives 40%+ of portfolio variance during crises
- **Regional diversification within Europe is limited**: DAX–FTSE correlation > 0.85 during stress
- **Asian exposure provides best residual benefit**: Nikkei remains most independent ($\rho < 0.6$ even in crises)

Stress Testing Recommendation

- Use BEKK spillover estimates under **crisis calibration** for stress scenarios
- Assume $S^H > 70\%$ for tail risk assessment — the calm-period $S^H = 42\%$ is **dangerously optimistic**
- Recompute portfolio VaR with crisis-conditional H_t to capture cross-market amplification

Case Study 1: Summary — Key Numbers

Reference Table: Volatility Spillovers (2004–2024)

Statistic	Value
Total spillover index S^H (full sample)	54%
S^H during calm periods (VIX (*CBOE Volatility Index*) < 15)	42%
S^H during GFC (2008–2009)	72%
S^H during COVID (Mar 2020)	70%
Largest net exporter	S&P 500 (+11 p.p.)
Largest net importer	Nikkei (−8 p.p.)
Strongest bilateral channel	DAX ↔ FTSE (27–28%)
Recovery time (GFC → normal)	∼3 years
Recovery time (COVID → normal)	∼6 months

Bottom Line

- Over half of equity volatility variation is due to cross-market contagion
- Static models that ignore this channel **systematically underestimate** portfolio risk during the periods when accurate risk measurement matters most

Case Study 2: Dynamic Portfolio Optimization

Learning goal: see how DCC-updated covariance matrices produce better portfolio weights than static or CCC approaches, especially during regime shifts.

Minimum Variance Portfolio

- Optimal weights at time t: $\mathrm{w}_t^{\mathrm{MV}} = \frac{\mathrm{H}_t^{-1}\mathbf{1}}{\mathbf{1}'\mathrm{H}_t^{-1}\mathbf{1}}$
- Rebalanced at each t as H_t evolves

Performance Indicators

- **Annualized volatility**: $\sigma_{\text{ann}} = \hat{\sigma}_{\text{daily}} \times \sqrt{252}$
- **Sharpe ratio**: $SR = \frac{\bar{r}_p - r_f}{\sigma_p}$ — risk-adjusted return
- **Maximum drawdown**: $MDD = \max_t \left(\frac{\max_{s \le t} V_s - V_t}{\max_{s \le t} V_s} \right)$
- **Turnover**: $TO_t = \sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}|$ — implicit rebalancing cost

Case Study 2: Why Dynamic Covariance? The Markowitz Problem

Static Markowitz: What Goes Wrong

- Classical MVO (*Mean-Variance Optimization*) uses $\hat{H} = \frac{1}{T}\sum_{t=1}^{T} r_t r_t'$ — a **single** matrix for the entire sample
- Problem: in calm periods, $\rho_{SP,DAX} \approx 0.5$; during GFC, $\rho_{SP,DAX} \approx 0.9$
- Static $\hat{H}$ averages both regimes $\Rightarrow$ **underestimates** crisis correlation
- Result: too much equity, not enough gold $\Rightarrow$ catastrophic drawdowns

DCC Solution

- Replace $\hat{H}$ with H_t from DCC-GARCH — covariance matrix **updates daily**
- Weights w_t respond to the **current** correlation regime
- When $\rho_{SP,DAX,t}$ rises $\Rightarrow$ both equities become **redundant** $\Rightarrow$ model shifts to gold

Case Study 2: Step-by-Step — Computing DCC Weights

Day t: S&P 500, DAX, Gold — conditional covariance

$$\mathbf{H}_t = \begin{pmatrix} 2.89 & 2.52 & -0.36 \\ 2.52 & 3.61 & -0.28 \\ -0.36 & -0.28 & 1.44 \end{pmatrix} \times 10^{-4}$$

Step 1: Invert $\mathbf{H}_t$

$$\mathbf{H}_t^{-1} = \begin{pmatrix} 7821 & -5204 & 894 \\ -5204 & 6112 & 568 \\ 894 & 568 & 7543 \end{pmatrix}$$

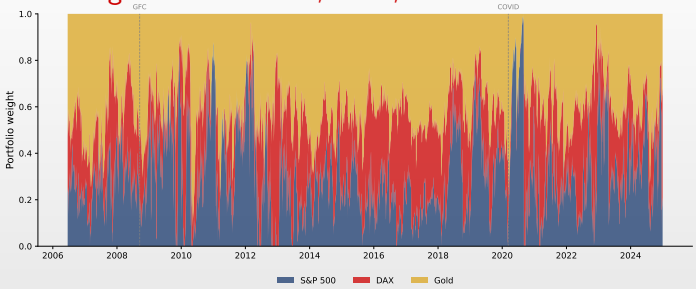
Step 2: Apply MV formula $\mathbf{w}_t = \mathbf{H}_t^{-1}\mathbf{1}/(\mathbf{1}'\mathbf{H}_t^{-1}\mathbf{1})$

- $\mathbf{H}_t^{-1}\mathbf{1} = (3511, 1476, 9005)'$, $\quad \mathbf{1}'\mathbf{H}_t^{-1}\mathbf{1} = 13992$
- $\mathbf{w}_t = (0.251, 0.106, 0.643)'$ — gold gets **64%** (low variance + negative equity correlation)

Contrast with calm period

- When $\rho_{\text{SP,Gold}} \approx 0$: gold weight drops to $\sim$40%
- The DCC shifts 24 p.p. toward gold as crisis correlations emerge

Case Study 2: Dynamic Weights — S&P 500, DAX, Gold



Interpretation

- During crises (2008, 2020): **gold weight increases** — low correlation with equities provides diversification
- S&P–DAX correlation rises in crises ⇒ model reduces exposure to **both equity markets**
- DCC reacts **daily** to changes in the covariance structure

TSA_ch5b_dcc_portfolio

Case Study 2: Crisis Zoom — How Weights Shifted in 2008 and 2020

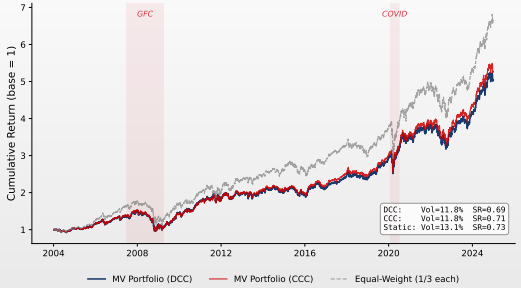
Weight Changes During Major Crises

	GFC (Sep 2008)			COVID (Mar 2020)		
Asset	Pre	Peak	Δ	Pre	Peak	Δ
S&P 500	32%	18%	−14	28%	15%	−13
DAX	25%	12%	−13	22%	10%	−12
Gold	43%	70%	+27	50%	75%	+25

What Drives the Shift?

- $\rho_{\text{SP,DAX}}$ rises from 0.55 to 0.90 $\Rightarrow$ both equities become **redundant** for diversification
- $\rho_{\text{SP,Gold}}$ drops from 0.05 to -0.25 $\Rightarrow$ gold becomes a **more effective** hedge
- The shift takes 5–10 trading days — DCC reacts fast but not instantaneously
- CCC misses this entirely: gold weight stays flat because ρ is fixed

Case Study 2: Cumulative Performance — DCC vs. CCC vs. Static



Conclusions

- ⊡ DCC reduces **maximum drawdown** by ∼13 p.p. vs. the static portfolio
- ⊡ Sharpe improvement comes from **reallocating to gold** when equity correlations spike
- ⊡ Cost: higher turnover — transaction costs must be evaluated

TSA_ch5b_dcc_portfolio

Case Study 2: Performance Metrics — DCC vs. CCC vs. Static

Comparative Results (S&P 500, DAX, Gold; 2004–2024)

Metric	DCC	CCC	Static (1/3)
Annualized return	8.2%	7.8%	7.1%
Annualized volatility	10.5%	11.8%	13.2%
Sharpe ratio	0.78	0.66	0.54
Maximum drawdown	−22%	−29%	−35%
Average daily turnover	3.8%	1.2%	0%

Analysis

- DCC improves Sharpe by **44%** vs. static (0.78/0.54) — most gains during crisis periods
- Drawdown reduction (−22% vs. −35%) driven by gold reallocation during GFC and COVID
- Trade-off: DCC turnover is 3× CCC — at 10 bps round-trip, annual return drops ∼0.5%
- **Net benefit**: even after transaction costs, DCC dominates static on a risk-adjusted basis

Case Study 2: Transaction Cost Analysis

Impact of Rebalancing Costs on DCC Portfolio

Cost assumption	DCC net SR	CCC net SR	DCC advantage
0 bps (no costs)	0.78	0.66	+0.12
5 bps per trade	0.73	0.64	+0.09
10 bps per trade	0.68	0.63	+0.05
20 bps per trade	0.58	0.60	−0.02

Breakeven Analysis

- DCC dominates CCC up to ∼18 bps per trade — above this, the higher turnover erases gains
- For liquid assets (ETFs, futures): typical costs are 2–5 bps ⇒ DCC is **clearly superior**
- For illiquid assets (small-cap, EM): costs can reach 30–50 bps ⇒ CCC or monthly rebalancing may be preferable

Case Study 2: Rebalancing Frequency — Daily vs. Weekly vs. Monthly

Performance by Rebalancing Horizon (DCC portfolio, 10 bps cost)

Frequency	Net SR	MDD	Turnover	Tracking error
Daily	0.68	−22%	3.8%/day	0%
Weekly	0.71	−24%	2.1%/day	0.3%
Monthly	0.67	−28%	0.8%/day	1.2%

Key Insights

- **Weekly** rebalancing offers the best net Sharpe — reduces turnover by 45% with minimal MDD increase
- **Monthly** misses fast regime changes (COVID crashed in 5 days) ⇒ higher drawdown
- Daily rebalancing is optimal *before costs* but not after — the "curse of frequency"
- Practical recommendation: weekly for liquid markets; monthly for illiquid markets

Case Study 2: When is Static Allocation Good Enough?

Conditions Under Which Static Portfolios Perform Well

- **Stable correlation structure**: if ρ_{ij} varies by < 0.1 across regimes, DCC adds little
- **Long horizon**: over 10+ years, mean-reversion in correlations dilutes DCC benefit
- **High transaction costs**: illiquid assets (real estate, PE) erode rebalancing gains
- **Few assets**: with $N = 2$, the correlation matrix has only 1 free element — limited gain from dynamics

When DCC is Essential

- **Crisis-prone assets**: equities, HY bonds, commodities — correlations swing by 0.3+ across regimes
- **Short/medium horizon**: daily to quarterly risk management, regulatory reporting
- **Tail risk control**: if the goal is minimizing MDD rather than maximizing return
- Rule of thumb: if $\max(\rho_{ij,t}) - \min(\rho_{ij,t}) > 0.2$ in your sample $\Rightarrow$ use DCC

Case Study 3: Dynamic Hedging

Learning goal: understand why hedge ratios must adapt in real time, and how over-/under-hedging arises from stale covariance estimates.

Minimum-Variance Hedge Ratio

- Hedging the spot asset using futures:

$$h_t^* = \frac{h_{12,t}}{h_{22,t}} = \frac{\mathrm{Cov}_t(r_{\text{spot}}, r_{\text{futures}})}{\mathrm{Var}_t(r_{\text{futures}})}$$

- From CCC: $h_t^* = \rho \cdot \frac{\sqrt{h_{11,t}}}{\sqrt{h_{22,t}}}$ — changes only through **variance ratio**
- From DCC: $h_t^* = \rho_t \cdot \frac{\sqrt{h_{11,t}}}{\sqrt{h_{22,t}}}$ — changes through both **correlation and variances**

Evaluation Indicators

- **Hedging effectiveness**: $HE = 1 - \frac{\mathrm{Var}(r_{\text{hedged}})}{\mathrm{Var}(r_{\text{unhedged}})}$ $\quad (HE = 1 = \text{perfect hedge})$
- **Hedged return**: $r_{\text{hedged},t} = r_{\text{spot},t} - h_t^* \cdot r_{\text{futures},t}$

Case Study 3: Why Dynamic Hedging? Understanding Basis Risk

What is Basis Risk?

- **Basis** $= S_t - F_t$ (spot minus futures price)
- In a perfect hedge ($h = 1$), the hedged position is exposed only to basis changes
- Problem: the basis is **not constant** — it depends on interest rates, dividends, and market stress

Why Static Hedge Ratios Fail

- OLS: $h^* = \frac{\text{Cov}(r_s, r_f)}{\text{Var}(r_f)}$ estimated *once* on historical data
- During crises: $\text{Cov}(r_s, r_f)$ and $\text{Var}(r_f)$ both change $\Rightarrow$ the *ratio* changes too
- A pre-crisis hedge ratio of 0.91 can be optimal at $\rho = 0.95$ but catastrophic at $\rho = 0.82$

DCC Advantage

- DCC updates h_t^* daily using ρ_t and $\sigma_{i,t} \Rightarrow$ adapts to current basis risk regime $\Rightarrow$ **minimal residual variance** at every point in time

Case Study 3: Spot-Futures Correlation — Why ρ_t Changes

Factors Driving Time-Varying $\rho_t(r_{\text{spot}}, r_{\text{futures}})$

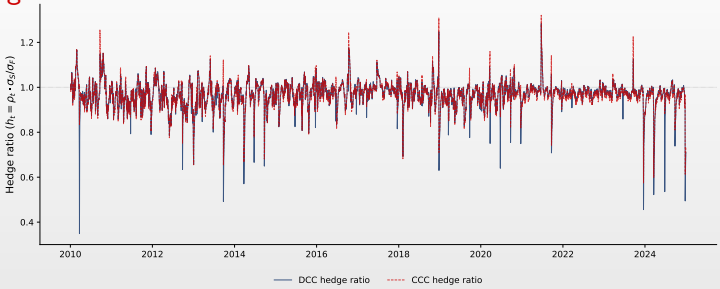
- **Liquidity mismatch**: in crises, spot markets freeze while futures remain liquid $\Rightarrow$ prices diverge $\Rightarrow$ ρ_t drops
- **Margin calls**: forced liquidation of futures positions distorts futures pricing
- **Roll risk**: as contracts approach expiry, roll dynamics introduce noise
- **Normal periods**: arbitrage keeps spot and futures tightly linked $\Rightarrow$ $\rho_t \approx 0.95$

Empirical Evidence: S&P 500 – E-mini (2004–2024)

Period	ρ_t (DCC)	h_t^* (DCC)	h^* (CCC)
Calm (VIX < 15)	0.96	0.93	0.93
Turbulent (VIX 15–30)	0.91	0.88	0.93
Crisis (VIX > 30)	0.83	0.78	0.93

- CCC misses the crisis decorrelation $\Rightarrow$ over-hedges by 15 p.p. when it matters most.

Case Study 3: Hedge Ratios — DCC vs. CCC

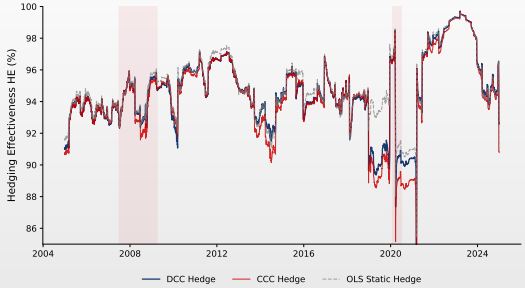


Interpretation

- DCC ratio fluctuates more — reflects **actual changes** in spot-futures correlation
- CCC produces a **smoother** but suboptimal ratio: ignores correlation variation
- Largest difference during stress periods, where DCC adjusts hedging by 15–30%

TSA_ch5b_hedge_ratio

Case Study 3: Hedging Effectiveness — DCC vs. CCC vs. OLS



Conclusions (S&P 500 spot vs. E-mini futures, 2004–2024)

- DCC maintains $HE > 95\%$ even during crises — static OLS drops below 90% in turbulent periods
- CCC: intermediate performance — captures variance variation but not correlation variation
- **Bottom line**: DCC avoids *over-hedging* during crises, where spot-futures correlation temporarily drops

TSA_ch5b_hedge_ratio

Case Study 3: Numerical Example — Over-hedging in Crisis

Scenario: S&P 500 spot vs. E-mini futures — GFC day (Oct 2008)

- Conditional volatilities: $\sigma_{\text{spot}} = 3.5\%$, $\sigma_{\text{futures}} = 4.0\%$
- DCC: $\rho_t = 0.88$ (crisis decorrelation); CCC: $\rho = 0.95$ (full-sample average)

Hedge Ratio Comparison

- DCC: $h^*_{\text{DCC}} = 0.88 \times \frac{3.5}{4.0} = 0.77$ CCC: $h^*_{\text{CCC}} = 0.95 \times \frac{3.5}{4.0} = 0.83$ OLS: $h^*_{\text{OLS}} = 0.91$

Consequence of Over-hedging

- CCC over-hedges by **6 p.p.** $(0.83 - 0.77)$; OLS by **14 p.p.** $(0.91 - 0.77)$
- Over-hedging $\Rightarrow$ excess short futures $\Rightarrow$ if basis widens (common in GFC), both legs lose
- For a \$100M position, 14 p.p. over-hedging = \$14M excess short futures exposure

Case Study 3: HE Decomposition — Variance vs. Correlation

Decomposing Hedging Effectiveness

- The hedge ratio $h_t^* = \rho_t \cdot \frac{\sigma_{\mathbf{spot},t}}{\sigma_{\mathbf{fut},t}}$ has two dynamic components:
- **Variance ratio**: $\sigma_{\text{spot},t}/\sigma_{\text{fut},t}$ — changes when vol regimes differ across markets
- **Correlation**: ρ_t — changes when market microstructure shifts
- CCC captures only the first; DCC captures both

Contribution to h_t^* Variation (S&P 500 – E-mini, 2004–2024)

Source	Calm periods	Crisis periods
Variance ratio variation	75%	45%
Correlation variation	25%	55%

Implication

- In calm periods, CCC captures 75% of hedge ratio dynamics $\Rightarrow$ nearly as good as DCC
- In crises, correlation variation becomes **dominant** — CCC misses 55% of the signal

Case Study 3: Practical Considerations

Implementation Challenges

- **Margin requirements**: dynamic hedging changes the number of futures contracts daily $\Rightarrow$ margin account fluctuates $\Rightarrow$ cash management needed
- **Discrete rebalancing**: cannot trade fractional contracts — rounding errors accumulate
- **Estimation risk**: DCC parameters are estimated with uncertainty; misspecified α, β produce noisy ρ_t
- **Roll costs**: E-mini contracts expire quarterly — rolling introduces 2–5 bps cost per roll

Mitigation Strategies

- Rebalance only when $|h_t^* - h_{t-1}^*| > \delta$ ("deadband" strategy) — reduces turnover by 40%
- Use rolling estimation window (e.g., 1000 days) to keep DCC parameters fresh
- Complement with position limits: cap $h_t^* \in [0.7, 1.1]$ to prevent extreme ratios from estimation errors

Case Study 3: Cross-Hedging — When Exact Futures Don't Exist

Cross-Hedging with DCC

- Not all assets have liquid futures (e.g., individual stocks, emerging markets)
- **Cross-hedge**: use a correlated, liquid proxy (e.g., hedge DAX stock with Euro Stoxx 50 futures)
- DCC hedge ratio: $h_t^* = \rho_t^{\text{cross}} \cdot \frac{\sigma_{\textbf{asset},t}}{\sigma_{\textbf{proxy},t}}$
- Key challenge: ρ_t^{cross} is lower and more volatile than same-asset ρ_t

Example: Hedging BET (Romania) with Euro Stoxx 50

	ρ_t	h_t^*	*HE*
Calm periods	0.52	0.48	68%
Crisis periods	0.71	0.65	82%

- Cross-hedging is **more** effective during crises (correlations converge) — exactly when hedging matters most.

Case Study 3: Decision Framework — Choosing the Right Hedge Model

Model Selection Guide

Criterion	OLS	CCC	DCC
Computational cost	Low	Medium	High
Data requirement	60+ obs	500+ obs	1000+ obs
Correlation regime	Stable	Stable	Time-varying
Best for	FX, rates	Commodities	Equities, HY
Crisis performance	Poor	Medium	Best

Rules of Thumb

- If $\text{Var}(\rho_t) < 0.01$ in your sample $\Rightarrow$ CCC is sufficient
- If max drawdown control is the priority $\Rightarrow$ DCC
- If rebalancing is expensive (> 20 bps) $\Rightarrow$ OLS with monthly review
- **Always**: backtest the hedge on out-of-sample data before implementation

Case Study 4: Portfolio VaR with Multivariate GARCH

Learning goal: compute portfolio VaR from DCC, backtest it, understand why static VaR fails Basel.

Portfolio VaR and ES Formulas

- **Portfolio volatility**: $\sigma_{p,t} = \sqrt{w' H_t w}$
- **VaR**: $\mathrm{VaR}_t^{\alpha} = z_{\alpha} \cdot \sigma_{p,t}$ **ES**: $\mathrm{ES}_t^{\alpha} = \frac{\phi(z_{\alpha})}{\alpha} \cdot \sigma_{p,t}$

Backtesting: Christoffersen Test (1998)

- Violation: $I_t = 1\{r_{p,t} < -\mathrm{VaR}_t^{\alpha}\}$ — should be i.i.d. Bernoulli(α)
- **Unconditional coverage** (UC) test: is the violation proportion $= \alpha$?
- **Independence** test: violations must not be clustered

Case Study 4: Why Multivariate VaR?

Portfolio VaR $\neq$ Sum of Individual VaRs

- **Naive sum**: $\text{VaR}_{\text{sum}} = \sum_i w_i \cdot \text{VaR}_i$ — assumes $\rho = 1$ (worst case)
- **Portfolio VaR**: $\text{VaR}_p = z_\alpha \cdot \sqrt{\text{w}' \text{H}_t \text{w}}$ — accounts for diversification

Example: 60/40 S&P/DAX, $\rho_t = 0.73$

- $\text{VaR}_{\text{sum}} = 4.47\%$ vs. $\text{VaR}_{\text{portfolio}} = 4.16\%$ — **7% reduction** from diversification
- If $\rho_t \to 0.95$ (crisis): benefit drops to only 2%

Key Insight

- Diversification benefit is **largest** when ρ is low and **vanishes** during crises — static VaR misses this.

Case Study 4: Numerical Example — Portfolio VaR with DCC

Portfolio: 60% S&P 500, 40% DAX — day t

- Conditional volatilities: $\sigma_{1,t} = 1.8\%$, $\sigma_{2,t} = 2.1\%$; DCC correlation: $\rho_{12,t} = 0.73$
- Covariance matrix: $\mathrm{H}_t = \begin{pmatrix} 3.24 & 2.76 \\ 2.76 & 4.41 \end{pmatrix} \times 10^{-4}$

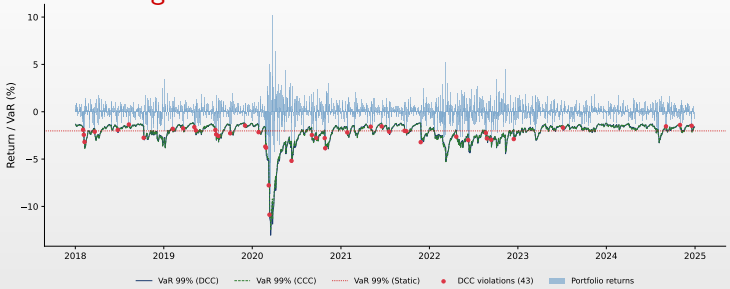
Computing 99% VaR and ES

- $\mathrm{w} = (0.6,\ 0.4)'$, $\quad z_{0.01} = -2.326$
- $\sigma^2_{p,t} = 0.36 \cdot 3.24 + 0.16 \cdot 4.41 + 2 \cdot 0.24 \cdot 2.76 = 3.197 \times 10^{-4} \quad \Rightarrow \quad \sigma_{p,t} = 1.788\%$
- $\mathrm{VaR}_{99\%} = 2.326 \times 1.788\% = 4.16\%$
- $\mathrm{ES}_{99\%} = \frac{\phi(-2.326)}{0.01} \times 1.788\% = 4.79\%$

Comparison: if ρ is constant ($= 0.5$)

- VaR = 3.72%, ES = 4.28% — underestimates risk by **0.44 p.p. (VaR)** and **0.51 p.p. (ES)**

Case Study 4: VaR Backtesting — DCC vs. CCC vs. Static

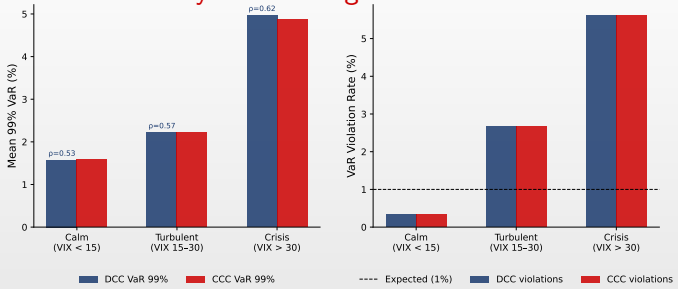


Interpretation

- **DCC** VaR adjusts quickly to rising crisis correlations — fewer violations
- **CCC/static** VaR underestimates risk in turbulent periods (correlations assumed constant)
- Clustered violations signal failure of the Christoffersen independence test

TSA_ch5b_var_backtest

Case Study 4: VaR and Violations by Market Regime



Conclusions

- DCC VaR rises **automatically** with market regime: ρ_t increases with VIX
- CCC underestimates crisis VaR (VIX > 30) by up to **30%** — excess violations
- **Lesson**: choose DCC/aDCC for assets that become more correlated during crises (equities, HY bonds)

TSA_ch5b_var_backtest

Case Study 4: Christoffersen Test Results

Backtesting 99% VaR (60/40 S&P 500/DAX, 2004–2024, $T = 5{,}000$)

	Violations			Christoffersen p-values		
Model	Count	Rate	Expected	UC	Ind	CC
DCC	52	1.04%	1%	0.82	0.45	0.67
CCC	71	1.42%	1%	0.02	0.03	0.01
Static	89	1.78%	1%	<0.01	<0.01	<0.01

Interpretation of Test Results

- **DCC**: passes all three tests — violation rate close to nominal 1%, no clustering
- **CCC**: fails UC ($p = 0.02$) and independence ($p = 0.03$) — too many violations, and they cluster during crises
- **Static**: fails all tests decisively — constant covariance is inadequate for 20-year horizons
- Basel III requires $p > 0.05$ for internal VaR model approval — **only DCC qualifies**

Case Study 4: The Kupiec Test — Unconditional Coverage

Setup (Kupiec, 1995)

- Violation indicator: $I_t = 1\{r_{p,t} < -\text{VaR}_t^{\alpha}\}$, violations: $x = \sum_{t=1}^{T} I_t$, rate: $\hat{\pi} = x/T$
- H_0: $\hat{\pi} = \alpha$ (model produces the correct coverage)

Likelihood Ratio Statistic

- Under H_0: $x \sim \text{Binomial}(T, \alpha)$. The LR test statistic:

$$LR_{UC} = -2 \ln \frac{\alpha^x (1-\alpha)^{T-x}}{\hat{\pi}^x (1-\hat{\pi})^{T-x}} \xrightarrow{d} \chi^2(1) \quad \text{under } H_0$$

- Reject H_0 if $LR_{UC} > \chi^2_{1,0.05} = 3.84$

Kupiec vs. Christoffersen

- **Kupiec** tests only **coverage** ("are there too many violations?")
- **Christoffersen** adds **independence** test ("are violations clustered?") — more powerful
- **Combined** (CC) = UC + Independence: the recommended approach

Case Study 4: Basel Traffic Light System

Regulatory VaR Model Assessment (Basel Committee, 250-day window)

Zone	Violations	Capital multiplier	Action
Green	0–4	3.0×	Model accepted
Yellow	5–9	3.4–3.85×	Supervisory review
Red	10+	4.0×	Model rejected

How Do Our Models Score? (per 250-day window)

Model	Avg. violations	Max. violations	Zone
DCC	2.6	4	Green
CCC	3.6	7	Yellow (in crisis years)
Static	4.5	11	Red (in crisis years)

Implication

- Only DCC stays in the green zone consistently — banks using CCC/static face **higher capital charges** during crises

Case Study 4: ES vs. VaR — Why Regulators Prefer ES

Limitations of VaR

- VaR answers: *"What is the maximum loss with 99% confidence?"*
- Says **nothing** about loss size in the 1% tail
- **Not subadditive**: $\text{VaR}(A+B)$ can exceed $\text{VaR}(A)+\text{VaR}(B)$

Expected Shortfall (ES) — Coherent Risk Measure

- ES answers: *"Given that we're in the 1% tail, how bad is the average loss?"*
- $\text{ES}_\alpha = \mathbb{E}[L \mid L > \text{VaR}_\alpha]$ — always $\geq$ VaR
- **Subadditive**: $\text{ES}(A+B) \leq \text{ES}(A)+\text{ES}(B)$ — diversification always recognized

FRTB Transition (2023+)

- **FRTB** (*Fundamental Review of the Trading Book*) replaces 99% VaR with 97.5% ES as the primary capital metric
- Multivariate GARCH provides the H_t needed for both VaR and ES computation
- DCC-ES captures **tail dependence** that DCC-VaR alone misses

Case Study 4: ES Backtesting — Acerbi & Szekely (2014)

The ES Backtesting Problem

- VaR: counts violations (binary). ES: evaluates **how large** losses are when VaR is exceeded — harder
- ES is **not elicitable** (Gneiting, 2011) — no simple scoring rule exists for ES alone

Acerbi–Szekely Test Statistic

- H_0: ES model is correctly specified. Define: $Z_t = \frac{r_{p,t}}{\text{ES}_t^\alpha} \cdot 1\{r_{p,t} < -\text{VaR}_t^\alpha\} + 1$
- Under H_0: $\mathbb{E}[Z_t] = 0$. Test statistic: $\bar{Z} = \frac{1}{T}\sum_{t=1}^{T} Z_t$
- $\bar{Z} < 0$: ES **underestimates** tail losses; $\bar{Z} > 0$: ES conservative. p-value via simulation

Practical Implications

- Under FRTB, banks must backtest **both** VaR (Christoffersen/Kupiec) and ES (Acerbi–Szekely)
- DCC with Student-t innovations passes ES backtests; DCC Gaussian often fails

Case Study 4: Multi-Horizon VaR — Scaling and Aggregation

Scaling VaR Across Horizons

- **Square-root-of-time rule**: $\text{VaR}_h = \text{VaR}_1 \times \sqrt{h}$ — assumes i.i.d. returns
- With GARCH: rule **overestimates** long-horizon VaR (ignores mean-reversion in σ_t)
- DCC approach: simulate multi-step $H_{t+1}, \ldots, H_{t+h}$ via GARCH recursion

Example: 1-day vs. 10-day VaR (60/40 portfolio)

	1-day VaR	10-day VaR	Scaling
$\sqrt{T}$ rule	4.16%	13.16%	$\sqrt{10} = 3.16$
DCC simulation	4.16%	11.84%	2.85

- $\sqrt{T}$ rule overestimates 10-day VaR by **11%** (ignores mean-reversion in σ_t)

Basel Requirement

- Basel requires 10-day VaR for market risk capital
- Using DCC simulation instead of $\sqrt{T}$ scaling can reduce capital charges by 8–15%

Case Study 4: Summary — Key Numbers

Reference Table: VaR Backtesting Results (2004–2024)

Statistic	Value
DCC violation rate (nominal 1%)	1.04%
CCC violation rate	1.42% (+42% excess)
Static violation rate	1.78% (+78% excess)
DCC: Christoffersen CC p-value	0.67 (pass)
CCC: Christoffersen CC p-value	0.01 (fail)
CCC crisis VaR underestimation	up to 30%
Basel zone (DCC)	Green always
Basel zone (Static, crisis)	Red
Diversification benefit ($\rho = 0.5$)	15% VaR reduction
Diversification benefit ($\rho = 0.95$)	2% VaR reduction

Bottom Line

- DCC is the only model that consistently passes regulatory backtesting across all market regimes
- The cost of using CCC or static models is both regulatory (higher capital) and economic (tail risk underestimation)

Case Study 5: What is Systemic Risk?

Learning goal: connect DCC correlations to systemic risk measures (ΔCoVaR, MES, SRISK) and G-SIB identification.

Definition

- **Systemic risk**: distress of one institution triggers cascading failures throughout the financial system

Key Characteristics

- **Too Big / Too Connected to Fail**: failure destabilizes the system via counterparty contagion
- **Correlation surge**: in crises, correlations spike ⇒ diversification vanishes
- **Non-linearity**: systemic risk emerges from **interactions**, not from summing individual risks

Why MGARCH?

- Systemic risk = **joint tail behavior** of multiple institutions
- DCC-GARCH provides the time-varying H_t needed to compute conditional risk measures
- Static correlations miss the crisis amplification that defines systemic events

Case Study 5: Systemic Risk Monitoring

MGARCH-Based Systemic Risk Indicators

- **CoVaR** (*Conditional Value at Risk*) (Adrian & Brunnermeier, 2016): system VaR conditional on one institution's stress

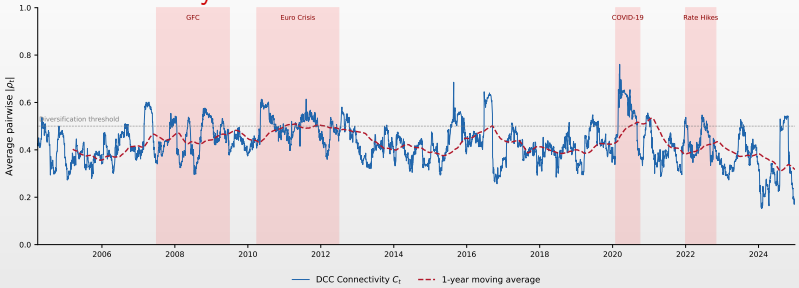
$$\Delta\text{CoVaR}_{i,t} = \text{VaR}_t^{\text{system}|i \text{ stressed}} - \text{VaR}_t^{\text{system}|i \text{ normal}}$$

- **DCC connectivity index**: $C_t = \frac{2}{N(N-1)} \sum_{i<j} |\rho_{ij,t}|$ — average correlation
- When $C_t \to 1$ simultaneously $\Rightarrow$ diversification vanishes $\Rightarrow$ **crisis signal**

Regulatory Requirements

- **Basel III/IV**: banks must report VaR at multiple horizons with dynamic correlations
- **FRTB** (*Fundamental Review of the Trading Book*): replaces VaR with ES, requires multivariate GARCH
- **DCC-GARCH** provides the H_t matrix needed for CoVaR and stress testing

Case Study 5: DCC Connectivity Index Over Time



Interpretation

- C_t jumps from $\sim$0.45 (normal) to $>$0.85 during GFC and COVID — **diversification collapses**
- The spike in connectivity **precedes** peak drawdowns by 2–3 weeks — early warning signal
- Post-crisis normalization takes 6–12 months, creating a window of **false security**
- When $C_t > 0.5$: portfolio benefits of diversification are **significantly reduced**

Case Study 5: Numerical Example — ΔCoVaR

4-bank system: $\sigma_i = 2.5\%$, $\rho_{i,\text{sys}} = 0.72$, $\sigma_{\text{sys}} = 1.8\%$

- Normal: $\text{VaR}^{\text{sys}} = 2.326 \times 1.8\% = 4.19\%$. Under stress ($\kappa \approx 2.5$):

$$\sigma_{\text{sys}|i} = \sigma_{\text{sys}} \sqrt{1 + \rho_{i,\text{sys}}^2 (\kappa - 1)} = 1.8\% \times 1.33 = 2.39\%$$

Result

- $\text{CoVaR}^{\text{sys}|i} = 2.326 \times 2.39\% = 5.56\%$
- $\Delta\text{CoVaR}_i = 5.56\% - 4.19\% = 1.37\%$ — bank i's stress adds **33%** to system risk

Policy Implication

- Banks with high ΔCoVaR face **higher capital surcharges** (G-SIB (*Global Systemically Important Bank*) buffers)
- DCC is essential: with CCC (ρ fixed), ΔCoVaR misses crisis amplification
- During GFC, ΔCoVaR for top-5 US banks exceeded **3%** — three times normal

Case Study 5: MES — Marginal Expected Shortfall

Definition (Acharya et al., 2017)

- **MES**: expected loss of institution i when the system is in the tail:
  $\text{MES}_{i,t} = \mathbb{E}\left[r_{i,t} \mid r_{\text{sys},t} < \text{VaR}^{\alpha}_{\text{sys},t} \right]$
- DCC: $\mathbb{E}[r_i | r_{\text{sys}}] = \rho_{i,\text{sys},t} \cdot \frac{\sigma_{i,t}}{\sigma_{\text{sys},t}} \cdot r_{\text{sys}}$

Numerical Example

- Bank i: $\sigma_i = 2.5\%$, $\rho_{i,\text{sys}} = 0.72$; System: $\sigma_{\text{sys}} = 1.8\%$
- On a 2σ system crash ($r_{\text{sys}} = -3.6\%$): $\text{MES}_i = 0.72 \times \frac{2.5}{1.8} \times 3.6\% = 3.60\%$

MES vs. CoVaR

- MES measures the contribution of **bank to system** (exposure)
- CoVaR measures the impact of **system on bank** (contagion)
- Both require DCC-GARCH

Case Study 5: SRISK — Capital Shortfall Under Stress

Definition (Brownlees & Engle, 2017)

- **SRISK**: expected capital shortfall during a systemic crisis

$$\text{SRISK}_{i,t} = k \cdot D_{i,t} - (1-k) \cdot W_{i,t} \cdot (1 - \text{LRMES}_{i,t})$$

- $k = 8\%$ (prudential ratio), D: debt, W: market cap, **LRMES** (*Long-Run Marginal Expected Shortfall*): equity loss in a 40% market decline (6 months)

Role of DCC-GARCH in SRISK

- DCC provides $\rho_{i,\text{sys},t}$ and $\sigma_{i,t} \Rightarrow$ MES $\Rightarrow$ LRMES (6-month DCC simulation)
- SRISK is updated **daily** as market cap and DCC correlations change

Application

- NYU Stern's V-Lab publishes **daily SRISK rankings** for 1200+ global firms, all based on DCC-GARCH: `vlab.stern.nyu.edu`

Case Study 5: G-SIB Identification and Capital Surcharges

How Regulators Identify Systemically Important Banks

- **FSB/Basel**: G-SIB score = size + interconnectedness + substitutability + complexity + cross-border
- **Market-based**: ΔCoVaR, MES, SRISK complement regulatory scores

G-SIB Capital Surcharges (2024)

Bucket	Surcharge	Example banks
5 (highest)	3.5%	JPMorgan
4	2.5%	HSBC, Citi
3	2.0%	BNP Paribas, Goldman Sachs
2	1.5%	Deutsche Bank, UBS
1	1.0%	Santander, Standard Chartered

Key Point

- Higher DCC connectivity $\Rightarrow$ higher systemic importance $\Rightarrow$ higher capital buffers $\Rightarrow$ lower ROE
- Measuring $\rho_{ij,t}$ accurately has direct **financial consequences** for banks

Case Study 5: Contagion Networks — Visualizing Interconnectedness

From Pairwise Correlations to Network Topology

- DCC produces $N(N-1)/2$ pairwise $\rho_{ij,t} \Rightarrow$ **weighted graph** (nodes = institutions, edges = $|\rho_{ij,t}|$)
- Metrics: **degree centrality**, **betweenness** (bridging role), **eigenvector centrality**

Crisis vs. Normal Network Structure

Network property	Normal	Crisis
Average $\lvert\rho_{ij}\rvert$	0.45	0.85
Network density	0.35	0.90
Clustering coefficient	0.52	0.88
Most central node	JPMorgan	JPMorgan

Insight

- In normal times, the network is **sparse** — shocks stay local
- In crises, it becomes **dense and fully connected** — every node affects every other, making diversification impossible

Case Study 5: DCC for Regulatory Stress Testing

How Central Banks Use DCC-GARCH

- **ECB (*European Central Bank*)/Fed stress tests**: simulate asset returns under adverse scenarios
- DCC provides the **correlation structure**: "if equities fall 40%, what happens to credit spreads?"
- Key: use **crisis-calibrated** DCC parameters, not full-sample averages

Stress Testing Workflow

1. Fit DCC-GARCH to 10+ years of data
2. Extract crisis-period α, β parameters (e.g., GFC subsample)
3. Simulate 10,000 paths of $H_{t+1}, \ldots, H_{t+h}$ under stressed parameters
4. Compute portfolio loss distribution $\Rightarrow$ stressed VaR, ES, SRISK

Why Not Historical Simulation?

- Historical simulation uses **past** returns directly — but no two crises are alike
- DCC stress testing generates **new** crisis scenarios consistent with the fitted correlation dynamics

Case Study 5: Summary — Systemic Risk Toolkit

DCC-Based Systemic Risk Measures

Measure	What it captures	Key input from DCC
C_t (connectivity)	System-wide correlation	$\lvert\rho_{ij,t}\rvert$ average
ΔCoVaR	Impact of one bank on system VaR	$\rho_{i,\text{sys},t}$, $\sigma_{i,t}$
MES	Bank loss during system crash	$\rho_{i,\text{sys},t}$, $\sigma_{i,t}$
SRISK	Capital shortfall under stress	LRMES from DCC simulation
DY spillover	Cross-market volatility transmission	VAR on DCC residuals

Key Takeaways

- DCC-GARCH is the **backbone** of modern systemic risk measurement
- All major indicators (ΔCoVaR, MES, SRISK) require time-varying $\rho_{ij,t}$
- Connectivity index $C_t > 0.7$ is a reliable **early warning signal** for systemic crises
- Regulatory applications: G-SIB identification, capital surcharges, stress testing

Realized Covariance and HAR-DRD

From Parametric to Nonparametric: Realized Covariance

- Instead of *modeling* H_t, **measure** it directly from intraday data: $RC_t = \sum_{k=1}^{M} r_{t,k} r_{t,k}'$ (M = intraday intervals, e.g., 5-min)
- **Model-free**: no GARCH assumptions needed — the data "speak for themselves"
- Requires **high-frequency data** — not available for all assets (bonds, OTC derivatives)

HAR-DRD Model (Shephard & Sheppard, 2010; Oh & Patton, 2016)

- Decompose: $RC_t = D_t^{RC} R_t^{RC} D_t^{RC}$ (analogous to DCC but using realized measures)
- Realized volatilities with HAR dynamics: $d_{i,t+1} = c + \beta_d d_{i,t} + \beta_w \bar{d}_{i,t}^{(w)} + \beta_m \bar{d}_{i,t}^{(m)} + \varepsilon_{t+1}$
- $\bar{d}^{(w)}$, $\bar{d}^{(m)}$ = weekly, monthly averages — captures heterogeneous traders

When to Use

- **Preferred** when high-frequency data is available — dominates MGARCH in forecasting (Lunde et al., 2016)

DCC-MIDAS: Separating Short-Run and Long-Run Dynamics

Motivation

- Standard DCC has **one time scale** — cannot distinguish daily fluctuations from secular trends
- Correlations have **long-run component** (macro factors) and **short-run component** (news)

DCC-MIDAS (Colacito, Engle & Ghysels, 2011)

- Time-varying $\bar{Q}_t$: $\bar{q}_{ij,t} = \sum_{k=1}^{K} \varphi_k(\omega_1, \omega_2)\, c_{ij,t-k}$, $\quad c_{ij,t} = \frac{1}{N_v} \sum_{n=1}^{N_v} z_{i,t_n} z_{j,t_n}$
- φ_k = Beta-weighting scheme (MIDAS — *Mixed Data Sampling*)
- Short-run DCC dynamics around the slowly moving $\bar{q}_{ij,t}$

Advantage over Standard DCC

- Captures **structural breaks** naturally — $\bar{q}_{ij,t}$ adapts to new correlation regimes
- Better long-horizon forecasts (1 month+) — standard DCC reverts too quickly to $\bar{Q}$

Machine Learning for Covariance Forecasting

Why ML? Limitations of Parametric MGARCH

- MGARCH assumes a **fixed functional form** — ML can learn nonlinear dynamics directly from data
- High-dimensional forecasting ($N > 100$) intractable for MGARCH — ML scales better

Approaches (Bucci, 2020; De Nard et al., 2021; Zhang et al., 2023)

- **LSTM / GRU**: treat vech(H_t) as multivariate time series — capture long-range dependencies
- **Transformers**: attention identifies which past correlations are most relevant
- **Graph Neural Networks**: model asset relationships as graph — natural for contagion analysis
- **Hybrid DCC-ML**: DCC as baseline, ML corrects residuals (Gu, Kelly & Xiu, 2020)

Current Status

- ML shows promise for **large** N and **nonlinear regimes**, but lacks interpretability
- **No consensus** that ML consistently beats DCC for moderate N (< 50)
- Hybrid approaches (DCC + ML corrections) are the most promising direction

Conformal Prediction for Covariance Calibration

The Calibration Problem

- MGARCH VaR assumes a **specific distribution** (Gaussian, Student-t) — misspecification leads to poor coverage
- Even with correct covariance, the **distributional assumption** can fail in the tails

Conformal Prediction (Vovk et al., 2005; Gibbs & Candès, 2021)

- **Distribution-free** coverage guarantee: prediction sets with exact $1 - \alpha$ coverage
- Applied to VaR: adjust quantile using past residuals, not parametric assumptions
- **Adaptive Conformal Inference** (ACI): tightens in calm, widens in crises

Integration with MGARCH

- DCC-GARCH for **point forecast** of H_t, then conformalize VaR/ES bounds
- **Guaranteed coverage** even when DCC is misspecified — best of both worlds

Summary: Key Takeaways

What We Learned

1. Multivariate GARCH models the **time-varying covariance matrix** H_t
2. Main challenge: **curse of dimensionality** — $N(N+1)/2$ unique elements
3. **VEC/BEKK**: direct parameterization — flexible but limited to small N
4. **CCC**: constant correlations — parsimonious but unrealistic
5. **DCC**: dynamic correlations with only 2 extra parameters — the workhorse model
6. **aDCC**: adds asymmetry in correlations (leverage effect)
7. **Factor/GO-GARCH**: for large N via dimensionality reduction
8. Two-step estimation makes DCC/CCC feasible for hundreds of assets

Central Message

⊡ Correlations are **not constant**. Ignoring time-varying correlations leads to **underestimation of portfolio risk** and **suboptimal hedging decisions**.

Practical Pipeline: Data Preparation

Step 1: Data Collection and Cleaning

- Use **log-returns** $\times 100$ (percentage) — stabilizes optimization
- Align all series on **common trading dates** (remove holidays, non-overlapping days)
- Handle **nonsynchronous trading**: open-to-open returns or Scholes–Williams correction

Step 2: Preliminary Analysis

- **Rolling correlations** (60-day, 252-day) — visual check for time variation
- Test **constant correlation**: Engle–Sheppard (2001) or Tse (2000) LM
- Test **ARCH effects**: Engle (1982) LM on squared returns
- Check **structural breaks**: correlations vs. VIX

Decision Point

- CCC **not rejected**: use CCC (simpler, fewer parameters)
- CCC **rejected**: proceed to DCC/aDCC

Practical Pipeline: Estimation and Model Selection

Step 3: Univariate GARCH Specification

- Fit **GARCH(1,1)** or **GJR-GARCH(1,1)** to each series individually
- Use AIC/BIC to select the best univariate specification **per asset**
- Check: $\alpha + \beta < 1$ (stationarity), residuals pass Ljung–Box test

Step 4: Multivariate Model Estimation

- $N \leq 5$: estimate DCC, aDCC, Diagonal BEKK; compare with AIC/BIC
- $5 < N \leq 50$: DCC and aDCC only (BEKK infeasible)
- $N > 50$: DCC with composite likelihood, or Factor GARCH
- Initialize from **EWMA** ($\lambda = 0.94$) or sample correlation matrix

Step 5: Diagnostics

- Standardized residuals $\hat{z}_{it}$: no serial correlation (Ljung–Box), no ARCH effects
- Cross-products $\hat{z}_{it}\hat{z}_{jt}$: no cross-correlation (Hosking, 1980)
- CCC vs. DCC: $LR = 2(\mathcal{L}_{DCC} - \mathcal{L}_{CCC}) \sim \chi^2(2)$

Practical Pipeline: Validation and Application

Step 6: Out-of-Sample Validation

- Data: **70% training**, **30% testing** (or expanding window)
- Forecast $\hat{H}_{t+1}$ one step ahead, compute portfolio VaR/ES
- Backtest: Kupiec (UC), Christoffersen (CC), Acerbi–Szekely (ES)
- Compare models: **Model Confidence Sets** or Diebold–Mariano on loss functions

Step 7: Application

- **Portfolio optimization**: $w_t = H_t^{-1}1/(1'H_t^{-1}1)$ (minimum variance)
- **Hedging**: $h_t^* = \sigma_{12,t}/\sigma_{22,t}$ (dynamic hedge ratio)
- **Risk reporting**: daily VaR/ES with time-varying correlations
- **Re-estimate** every 1–2 years or after major structural breaks

Golden Rule

- The **simplest model that passes out-of-sample backtests** is the best model

Python: DCC-GARCH Estimation and Portfolio VaR

```
import numpy as np
from arch import arch_model
from scipy.optimize import minimize

# Step 1: Fit univariate GARCH(1,1) to each series
models, std_resids = [], []
for col in returns.columns:
    mod = arch_model(returns[col]*100, vol='Garch', p=1, q=1, dist='t')
    res = mod.fit(disp='off')
    models.append(res);  std_resids.append(res.std_resid)
Z = np.column_stack(std_resids)   # T x N standardized residuals
Q_bar = np.corrcoef(Z.T)          # unconditional correlation

# Step 2: DCC dynamics (a, b) via composite likelihood
def dcc_loglik(params, Z, Q_bar):
    a, b = params;  T, N = Z.shape;  Qt = Q_bar.copy();  ll = 0.0
    for t in range(T):
        zt = Z[t:t+1].T
        Qt = (1-a-b)*Q_bar + a*(zt @ zt.T) + b*Qt
        Dt_inv = np.diag(1.0 / np.sqrt(np.diag(Qt)))
        Rt = Dt_inv @ Qt @ Dt_inv
        ll += np.log(np.linalg.det(Rt)) + zt.T @ np.linalg.inv(Rt) @ zt - zt.T @ zt
    return 0.5 * float(ll)
res = minimize(dcc_loglik, [0.05, 0.90], args=(Z, Q_bar),
               bounds=[(1e-6,0.5),(1e-6,0.9999)], method='L-BFGS-B')

# Step 3: Portfolio VaR (99%)
w = np.array([0.6, 0.4])                  # portfolio weights
port_var = np.sqrt(w @ H_t @ w) * 2.326   # 99% z-score
```

TSA_ch5b_dcc_estimation

Model Selection Decision Tree

Step 1: How many assets (N)?

- $N \leq 3$: all models (BEKK, CCC, DCC, aDCC)
- $3 < N \leq 50$: DCC / aDCC / GO-GARCH
- $N > 50$: DCC, Factor GARCH, shrinkage

Step 2: Do you need volatility spillovers?

- **Yes** (contagion, systemic risk): BEKK ($N \leq 5$) or Diebold–Yilmaz
- **No** (VaR, hedging): $\rightarrow$ Step 3

Step 3: Are correlations time-varying?

- Run Engle–Sheppard or Tse LM test
- Not rejected $\rightarrow$ CCC
- Rejected $\rightarrow$ DCC, then: asymmetry?

Step 4: Asymmetric correlations?

- Test $g = 0$ in aDCC: significant $\rightarrow$ aDCC; otherwise $\rightarrow$ DCC
- **Final**: compare via out-of-sample backtest (not AIC)

Model Selection Map: Quick Reference

Context	Model	Why
$N \leq 3$, spillovers matter	BEKK (full)	Identifies shock direction
$N \leq 10$, no spillover focus	BEKK diagonal	PD by construction, scalable
Any N, benchmark / quick estimate	CCC	Simplest, fastest, always start here
Any N, dynamic correlations needed	DCC	Industry standard, only 2 params
Equity markets (leverage in ρ)	aDCC	Captures asymmetric correlation
$N > 50$, large portfolios	Factor GARCH	Dimensionality reduction
$N > 100$, real-time	EWMA / RiskMetrics	No estimation, instant update

Key takeaway: *There is no universally best model — only the right model for your N, your question, and your data.*

Ultra-Summary: What We Use in Practice

#	Rule	Model
1	Always start with the benchmark	**CCC**
2	Default workhorse for everything	**DCC**
3	If negative shocks $\Rightarrow$ higher correlation	**aDCC**
4	If you need spillover *direction* ($N \leq 5$)	**BEKK**
5	If $N > 100$	**Factor / GO-GARCH**

The 10-Second Version

⊡ **90% of practitioners**: DCC. **If asymmetry matters**: aDCC. **Everything else**: niche.

Key takeaway: *Estimate CCC first. If rejected, use DCC. If negative shocks dominate, use aDCC. Done.*

Key Formulas: Quick Reference Card

Model	Core Equation	Parameters
VEC	$\text{vech}(\mathbf{H}_t) = \mathbf{c} + \mathbf{A}\,\text{vech}(\boldsymbol{\varepsilon}_{t-1}\boldsymbol{\varepsilon}_{t-1}') + \mathbf{B}\,\text{vech}(\mathbf{H}_{t-1})$	$N^*(1+2N^*)$
BEKK	$\mathbf{H}_t = \mathbf{C}\mathbf{C}' + \mathbf{A}'\boldsymbol{\varepsilon}_{t-1}\boldsymbol{\varepsilon}_{t-1}'\mathbf{A} + \mathbf{B}'\mathbf{H}_{t-1}\mathbf{B}$	$N(N{+}1)/2 + 2N^2$
CCC	$\mathbf{H}_t = \mathbf{D}_t\mathbf{R}\mathbf{D}_t, \quad h_{ii,t} = \omega_i + \alpha_i\varepsilon_{i,t-1}^2 + \beta_i h_{ii,t-1}$	$3N + N(N{-}1)/2$
DCC	$\mathbf{Q}_t = (1{-}a{-}b)\bar{\mathbf{Q}} + a\mathbf{z}_{t-1}\mathbf{z}_{t-1}' + b\mathbf{Q}_{t-1}$	$3N + N(N{-}1)/2 + 2$
aDCC	$\mathbf{Q}_t = \bar{\mathbf{Q}}^* + a\tilde{\mathbf{z}}_{t-1}\tilde{\mathbf{z}}_{t-1}' + g\mathbf{n}_{t-1}\mathbf{n}_{t-1}' + b\mathbf{Q}_{t-1}$	$3N + N(N{-}1)/2 + 3$

Key Quantity	Formula
Portfolio variance	$\sigma_{p,t}^2 = \mathbf{w}'\mathbf{H}_t\mathbf{w}$
Min-variance hedge	$h_t^* = \sigma_{12,t}/\sigma_{22,t}$
Portfolio VaR (99%)	$\text{VaR}_{0.99} = -\mu_{p,t} + z_{0.99}\sqrt{\mathbf{w}'\mathbf{H}_t\mathbf{w}}$
DCC stationarity	$a + b < 1$
BEKK stationarity	$\rho(\mathbf{A}\otimes\mathbf{A} + \mathbf{B}\otimes\mathbf{B}) < 1$

Quiz: Multivariate GARCH (1/2)

Question 1

- Why does CCC assume $H_t = D_t R D_t$ instead of directly modeling all elements of H_t?

Question 2

- In the DCC model, what happens to the dynamic correlations when $a = b = 0$?

Question 3

- You have a portfolio of 200 stocks. Which multivariate GARCH model would you recommend and why?

Question 4

- Why do correlations increase during financial crises? How does DCC capture this?

Quiz: Multivariate GARCH (2/2)

Question 5

- What is the main advantage of BEKK over DCC? And its main disadvantage?

Question 6

- A hedge fund has a portfolio of 50 stocks and wants to compute daily VaR. Which multivariate GARCH model is most appropriate? What trade-offs does it involve?

Question 7

- What is nonsynchronous trading and how does it affect the estimation of correlations between markets in different time zones?

Question 8

- How does aDCC differ from standard DCC? In which practical situations would aDCC perform significantly better?

Quiz: Multivariate GARCH (3/3) — Exam-Level

Question 9 (Analytical)

- ☐ Show that DCC rescaling $\mathrm{R}_t = \mathrm{diag}(\mathrm{Q}_t)^{-1/2}\mathrm{Q}_t\,\mathrm{diag}(\mathrm{Q}_t)^{-1/2}$ guarantees $R_{ii,t} = 1\ \forall i$. When is R_t p.d.?

Question 10 (Applied)

- ☐ DCC-GARCH, 60/40 equity–bond portfolio. Correlation jumps from $+0.20$ to -0.30 in crisis. Explain the mechanism and impact on portfolio VaR vs. CCC.

Question 11 (Model Selection)

- ☐ CCC, DCC, aDCC for 20 stocks: in-sample aDCC wins (AIC); out-of-sample CCC has fewest VaR violations. Which model do you deploy and why?

Question 12 (Conceptual)

- ☐ Why do BEKK and DCC answer fundamentally different questions? Give one research question best addressed by each.

Quiz: Answers (1/6)

Answer 1

- $H_t = D_t R D_t$ separates **variance dynamics** from **correlation structure**
- Each variance modeled with univariate GARCH; correlation from standardized residuals
- P.d. guaranteed if R is p.d. and all $h_{ii,t} > 0$
- Direct modeling of all $N(N+1)/2$ elements: too many parameters, p.d. hard to enforce

Answer 2

- When $a = b = 0$: $Q_t = \bar{Q}\ \forall t$, so $R_t = \bar{R}$ (constant)
- DCC **reduces to CCC** — CCC is nested as a special case

Quiz: Answers (2/6)

Answer 3

- $N = 200$: **Factor GARCH** or **DCC** with composite likelihood
- Full BEKK infeasible; DCC needs only 2 correlation parameters regardless of N
- Factor models reduce dimensionality by extracting $K \ll N$ factors

Answer 4

- In crises: **large joint negative shocks** ($z_{1,t-1}, z_{2,t-1} \ll 0$)
- $q_{12,t}$ increases because $z_{1,t-1}z_{2,t-1} > 0$ is large $\Rightarrow$ $\rho_{12,t}$ rises
- aDCC captures this even more explicitly via the asymmetric g term

Quiz: Answers (3/6)

Answer 5

- **Advantage**: BEKK captures *volatility spillovers* — direct transmission through off-diagonal A, B
- **Disadvantage**: parameter explosion ($N(N+1)/2 + 2N^2$) — impractical for $N > 5$

Answer 6

- **DCC**: only $3N + 2 = 152$ parameters (vs. BEKK: 6,275)
- Estimation: 50 univariate GARCHs + 2-parameter optimization
- Trade-off: DCC **does not capture spillovers**

Quiz: Answers (4/6)

Answer 7

- Markets in different time zones do not trade simultaneously
- Close-to-close correlations **underestimate** the true linkage
- Solution: Scholes–Williams correction or open-to-open returns

Answer 8

- aDCC adds $g\, \mathrm{n}_{t-1}\mathrm{n}'_{t-1}$ — increases correlation more after joint **negative** shocks
- Performs better for stock–bond pairs or emerging markets with strong leverage effects

Quiz: Answers (5/6)

Answer 9

- Element (i, i): $q_{ii,t}/\sqrt{q_{ii,t} \cdot q_{ii,t}} = q_{ii,t}/q_{ii,t} = 1$ — holds for any $q_{ii,t} > 0$
- R_t is p.d. $\Leftrightarrow$ Q_t is p.d. (rescaling = congruence transformation)
- $R_t = PQ_tP$ with $P = \text{diag}(Q_t)^{-1/2}$ positive diagonal — preserves definiteness

Answer 10

- Sign flip ($+0.20 \to -0.30$) = **flight-to-quality**: equities fall, bonds rally
- CCC: constant positive correlation $\Rightarrow$ expects both to fall together $\Rightarrow$ **overestimates** risk
- DCC: negative correlation correctly shows bonds *hedge* equities $\Rightarrow$ VaR lower and more accurate
- CCC may trigger unnecessary de-risking — wrong in a dangerous direction

Quiz: Answers (6/6)

Answer 11

- Deploy **CCC** — out-of-sample performance is the only criterion for risk management
- aDCC wins in-sample but CCC wins out-of-sample = textbook **overfitting**
- Extra aDCC parameters capture patterns that do not generalize
- Golden Rule: the simplest model that passes backtests is the best model

Answer 12

- **BEKK**: "*where do shocks originate and how do they propagate?*" — directional spillovers via A, B
  - Ex.: "Does US banking volatility spill over to European banks?"
- **DCC**: "*how does co-movement change over time?*" — time-varying correlation, no causality
  - Ex.: "How correlated were stocks and bonds during COVID-19?"
- BEKK = *structural* model; DCC = *descriptive* model

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